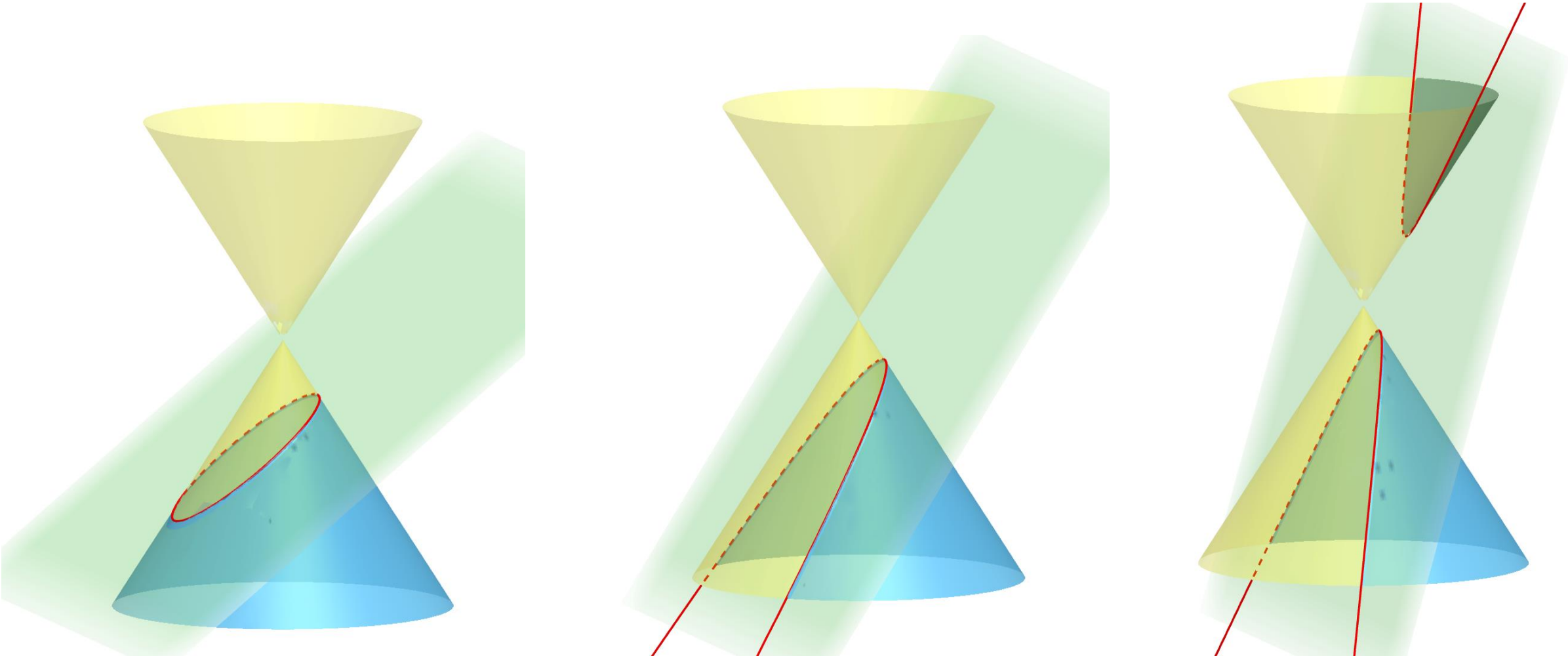
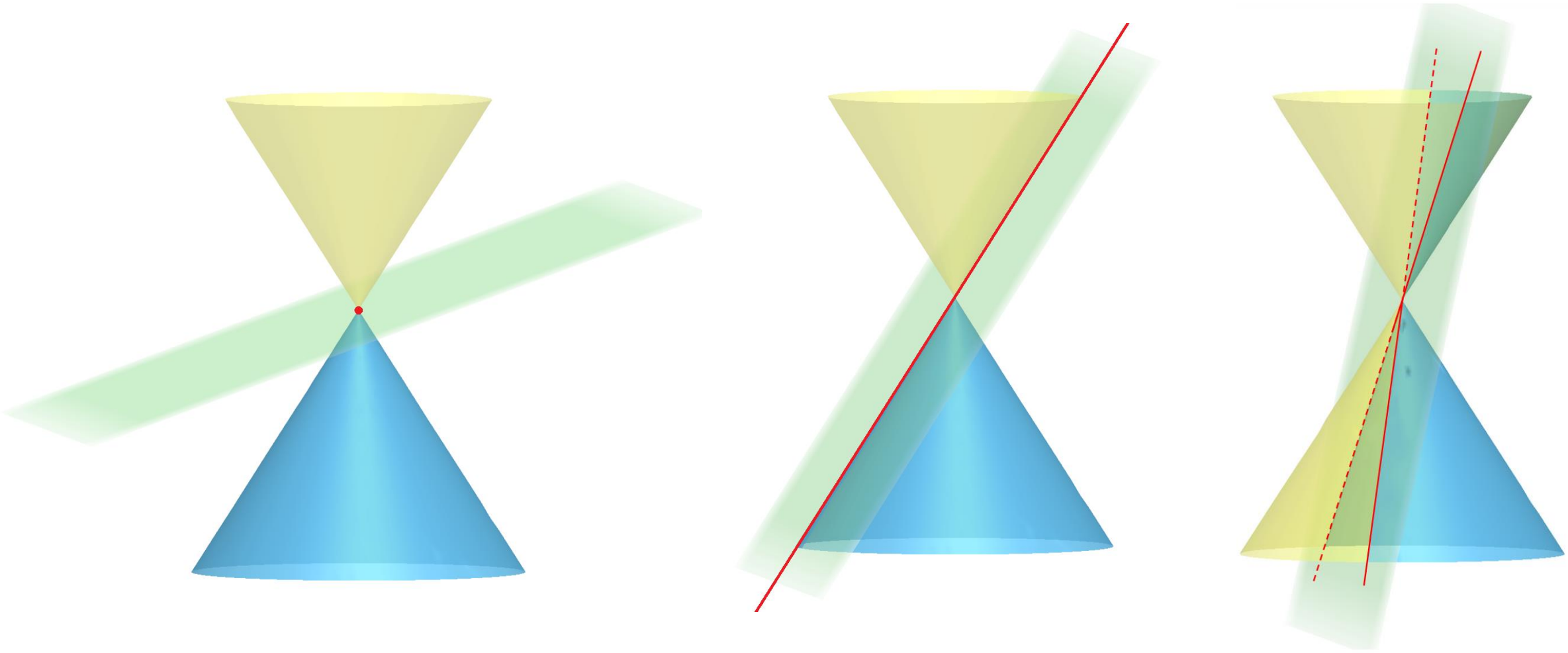
1.1.2 Section plane avec une lampe de poche

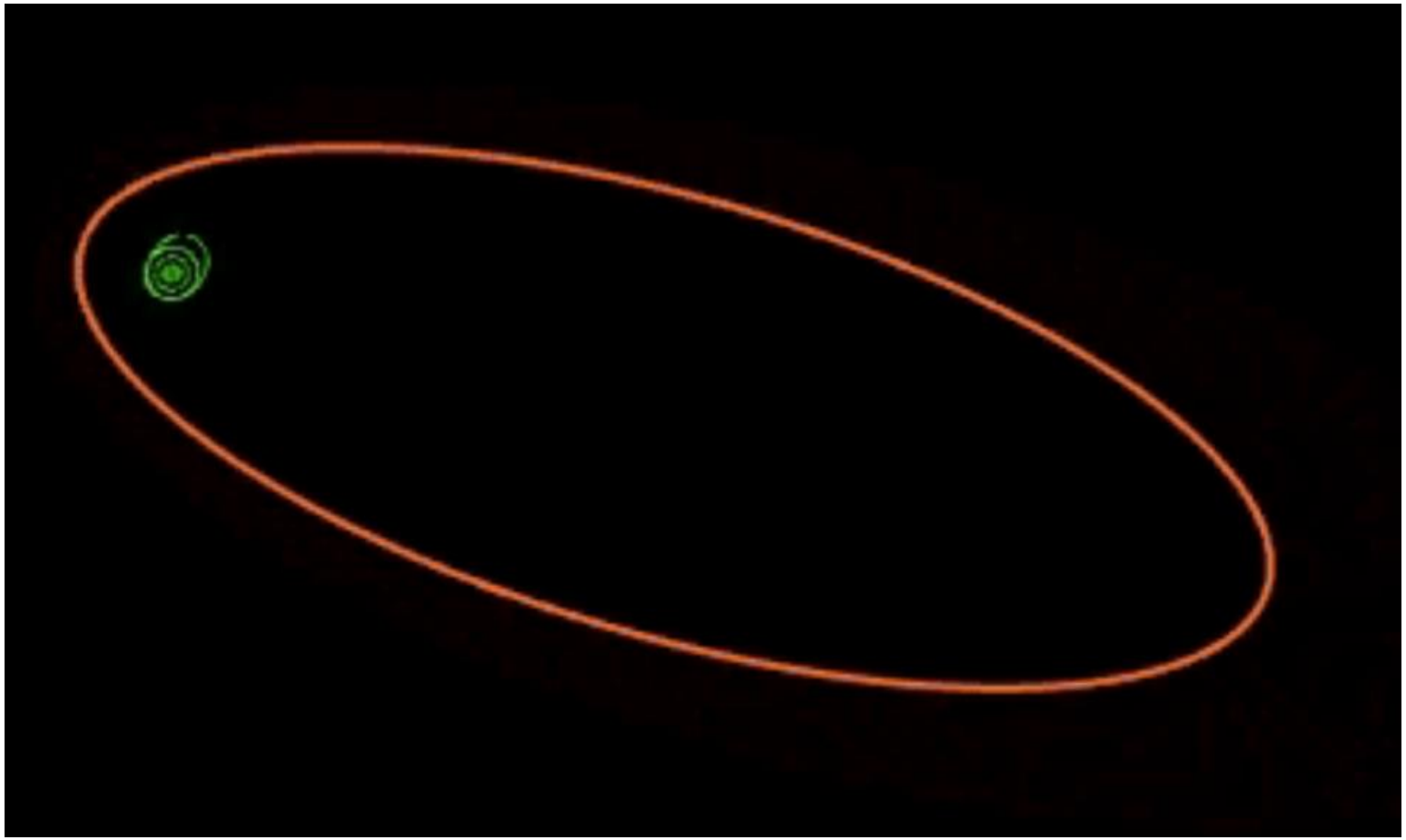
À l'aide d'un carton et d'une lampe de poche, créez différentes formes en deux dimensions sur le carton.

1. Combien de formes différentes pouvez-vous obtenir ? Dessinez-les.
2. Quelle est la forme tridimensionnelle du faisceau lumineux émis par la lampe de poche ?
3. Quels éléments influencent la forme obtenue sur le carton ?

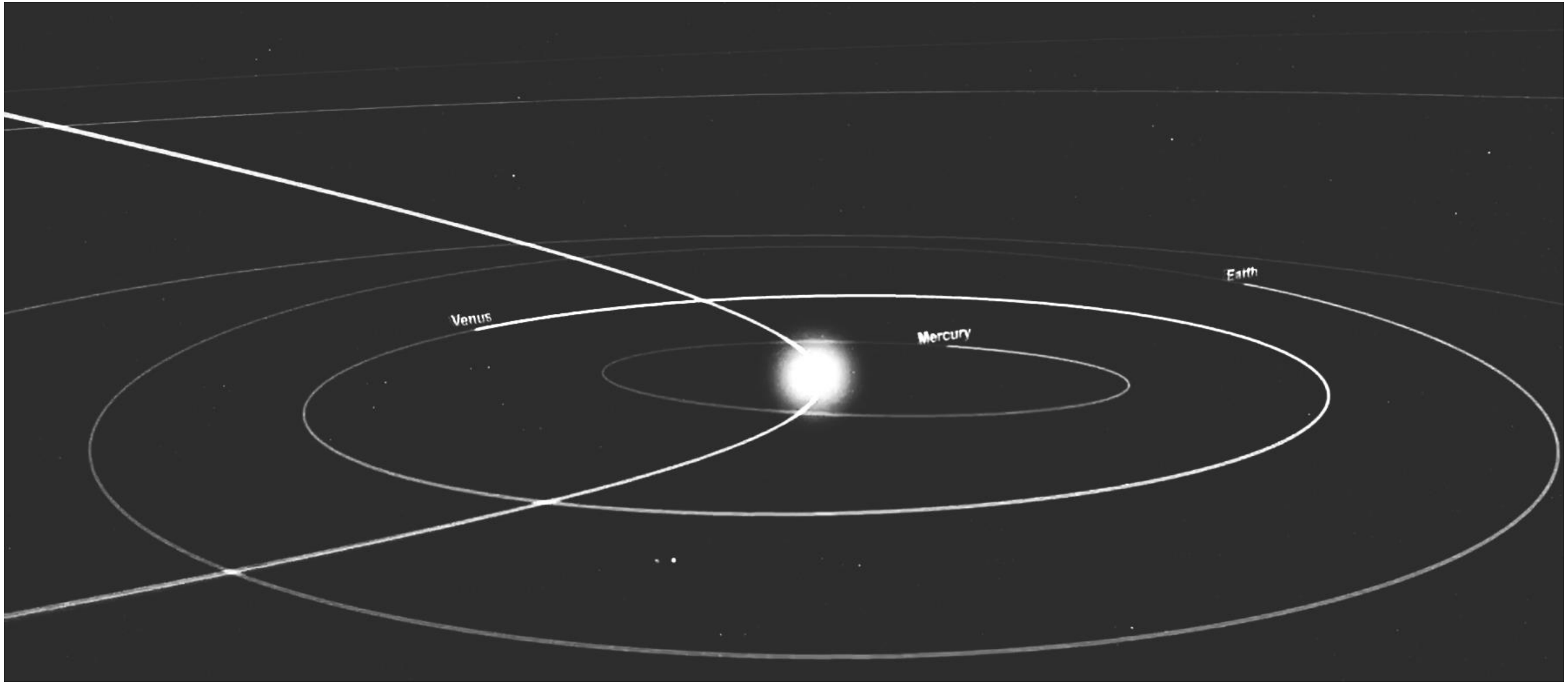
https://www.geogebra.org/classic/dt4yx9xv





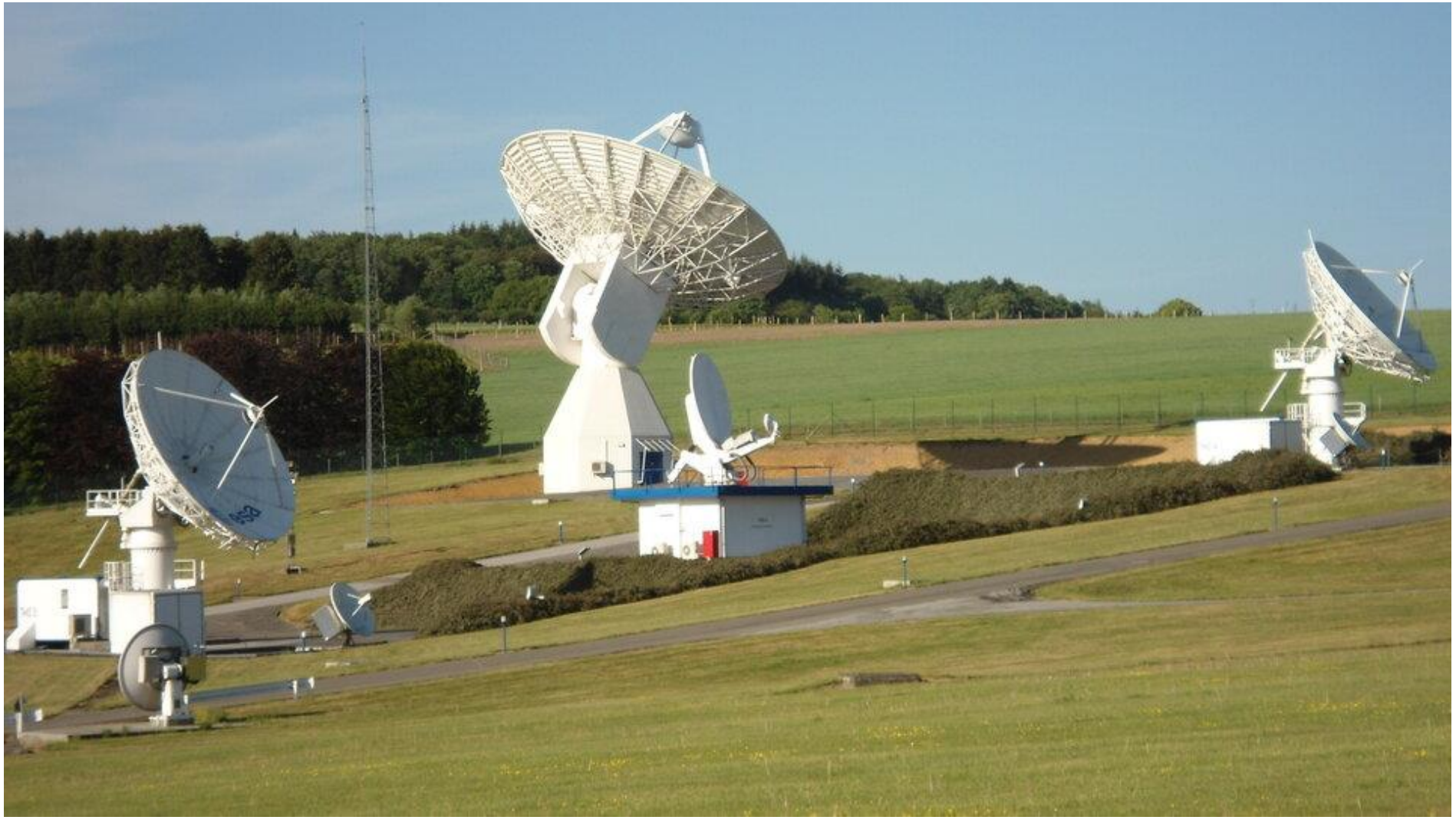


Trajectoire de Sedna, NASA

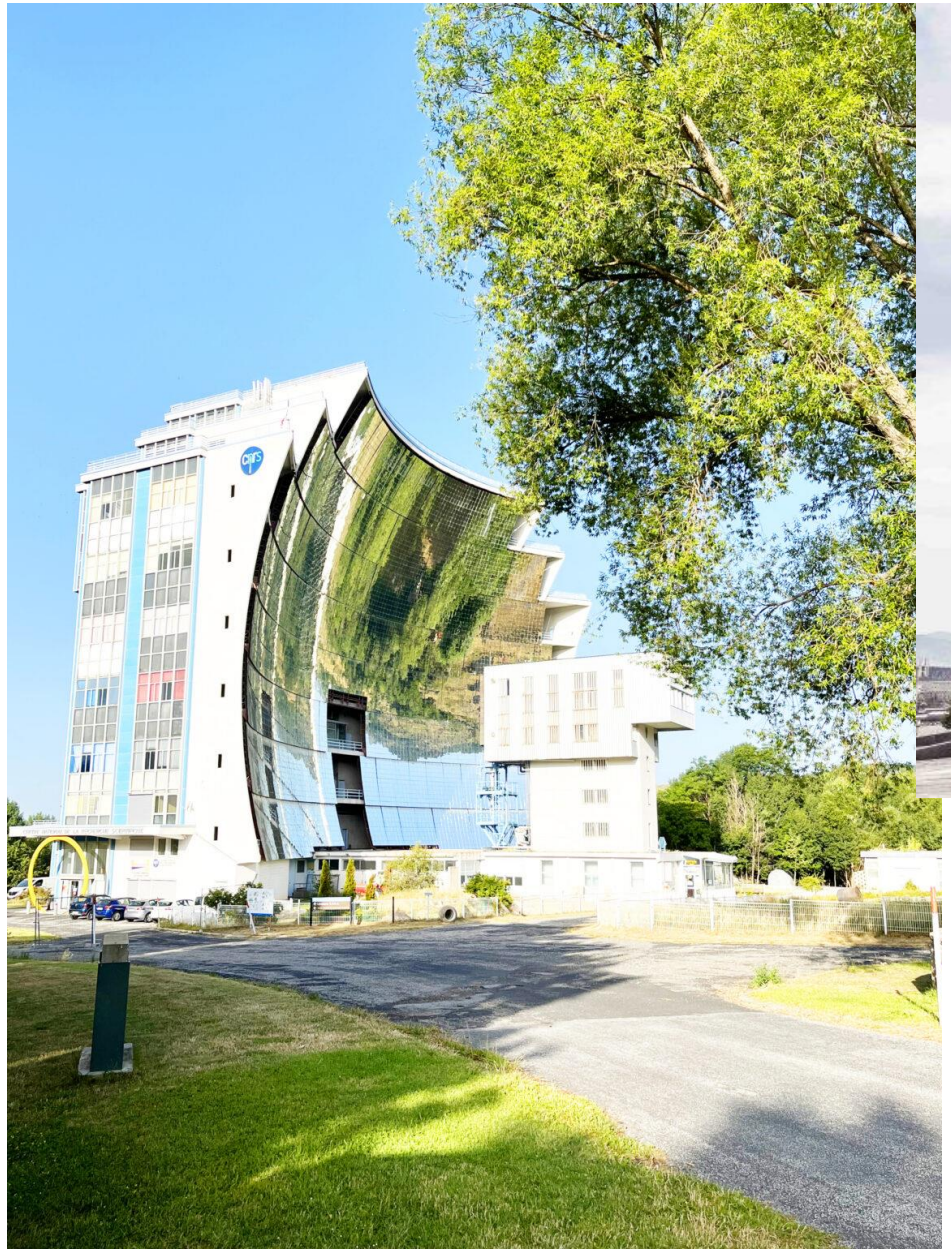


Trajectoire de la comète C/2012 S1 (ISON), NASA

www.youtube.com/watch?v=40wICUY5VmU



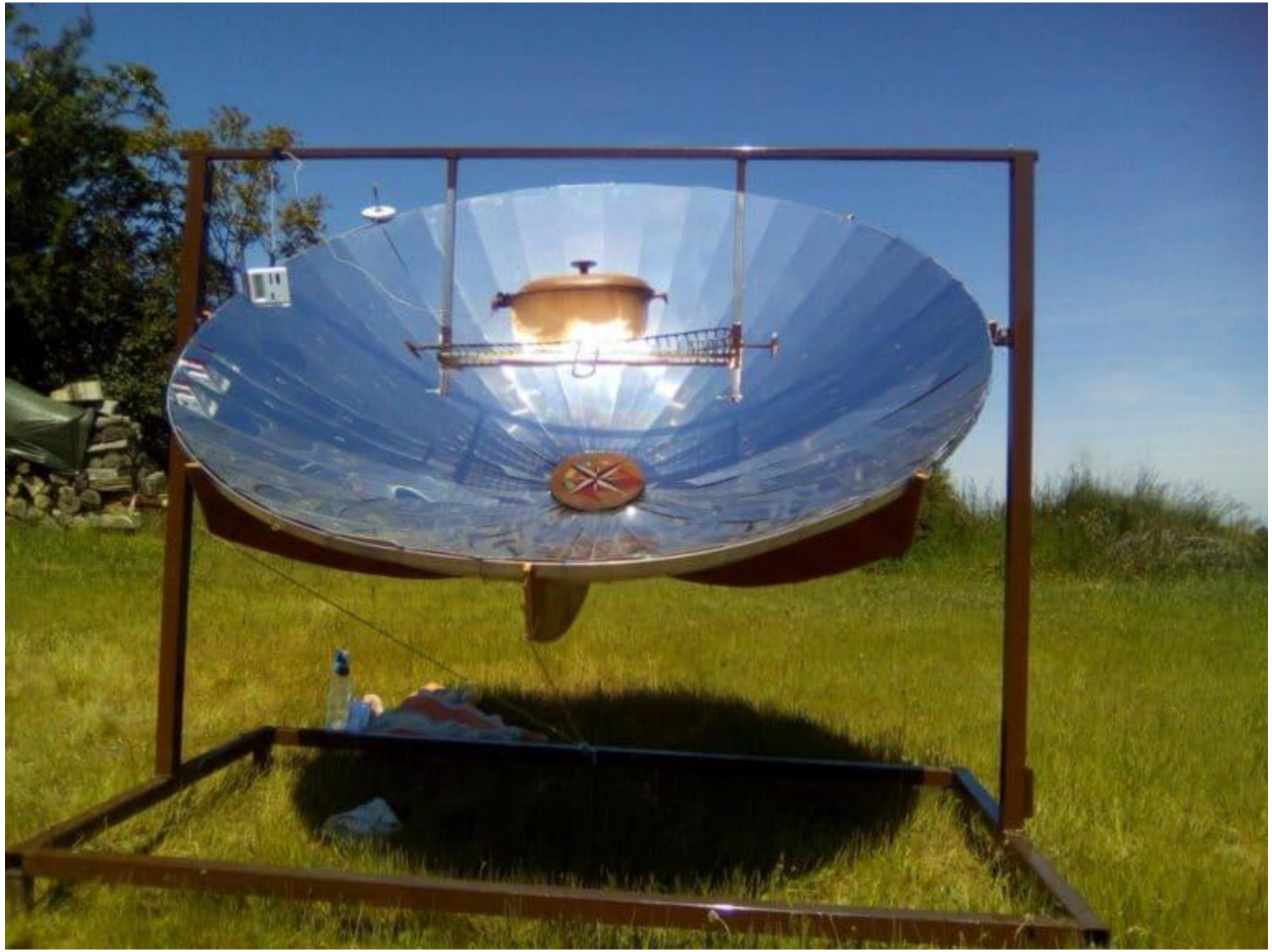
ESA, Redu, Belgique



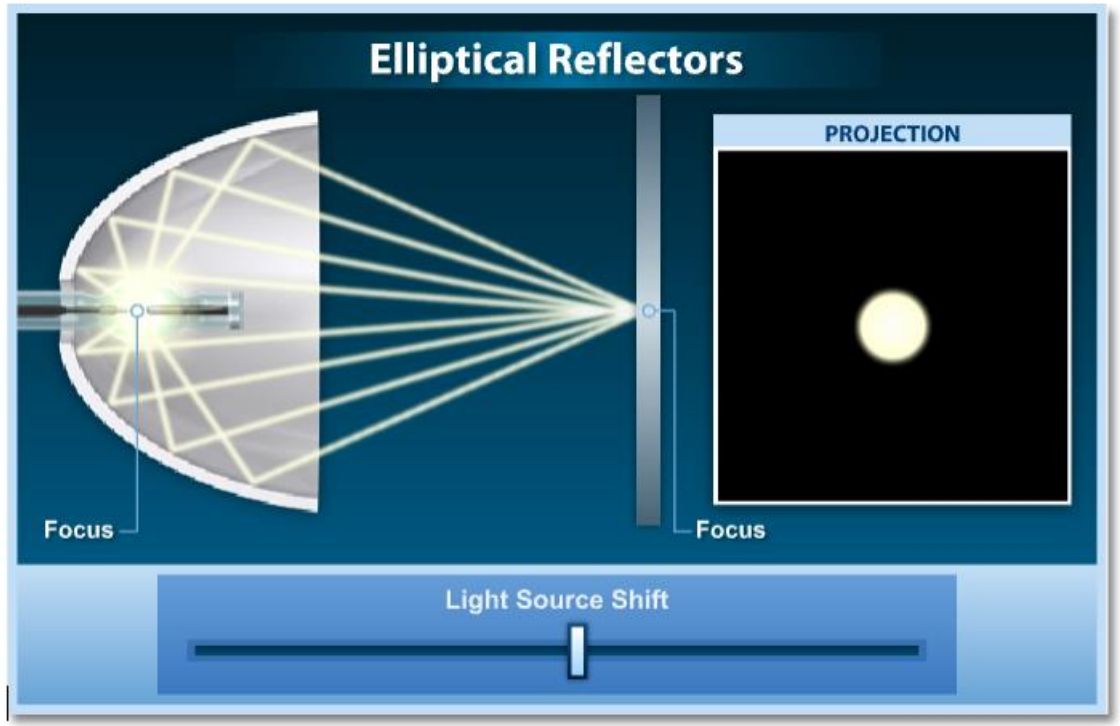
Institute of materials science Uzbekistan academy of science, https://imssolar.uz/en

Fours solaires (France et Ouzbékistan)

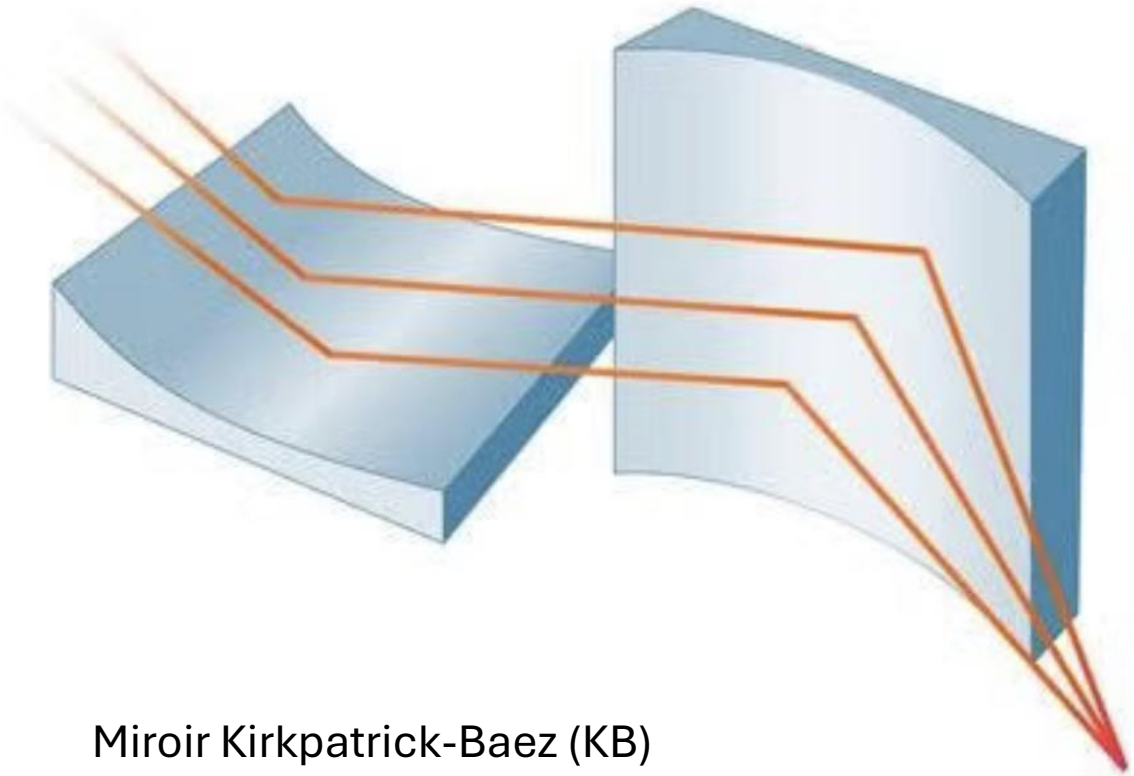
PROMES CNRS, https://www.promes.cnrs.fr



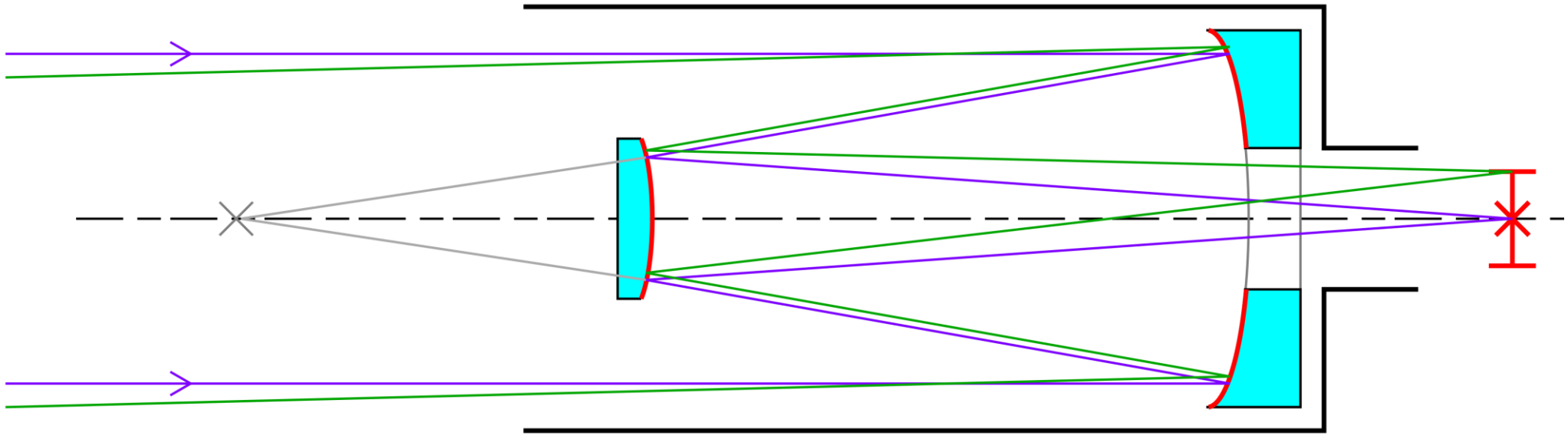
Atelier du zephyr, https://atelierduzephyr.org/auto-construction/cuisiner/cuiseur-et-four-solaires



Réflecteur elliptique dans une lampe pour microscope à fluorescence



Miroir Kirkpatrick-Baez (KB)
Pour les rayons X



Télescope cassegrain avec un miroir hyperbolique et un miroir parabolique

Télescope Hubble avec 2 miroirs hyperboliques

Tours de refroidissement de la centrale électrique de Tihange en forme d'hyperboloïde

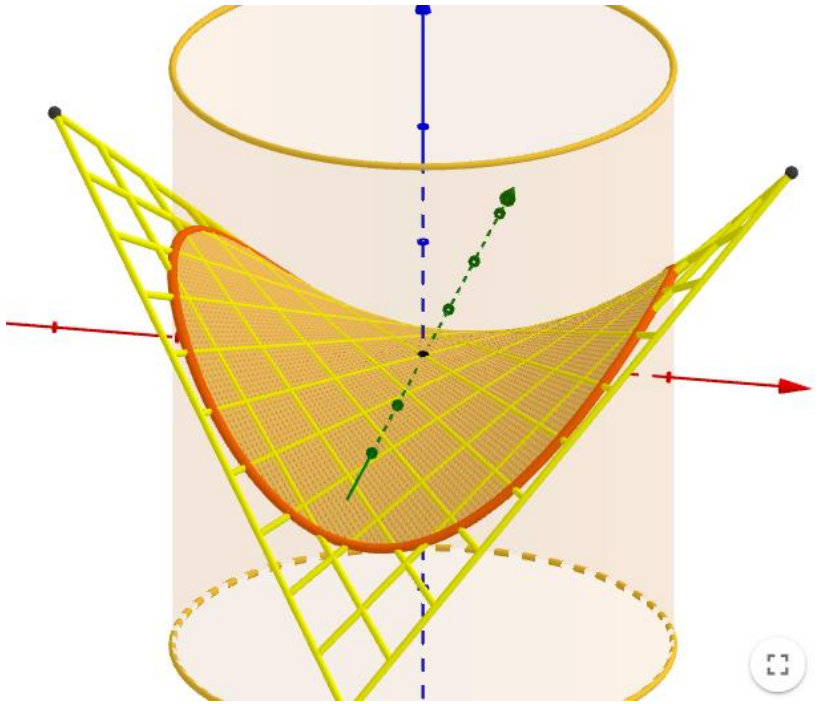
Gare en forme de paraboloïde hyperbolique à Varsovie

PRODUCTS COUNTRIES/REGIONS CONTACT US WHAT'S NEW BIG GAME EN ESPAÑOL

ALL ABOUT PRINGLES®

1. How are Pringles® crisps made?

Pringles® crisps are made with real dried potato flakes that we form into a 'hyperbolic paraboloid shaped potato dough'. We fry and season the crisps just right before placing them into our iconic can. The seasoning system we use is a waterfall process (seasoning gets sprinkled on) vs a tumbler like other chip processes.



Vincent Pantaloni , https://www.geogebra.org/m/jayhhbbs

Dans $\mathbb{R}^2$:

$$ax + by + c = 0,$$

$$ax^2 + bx + c + dy = 0, \quad \Leftrightarrow \quad y = ax^2 + bx + c.$$

Dans $\mathbb{R}^3$:

$$ax + by + cz + d = 0,$$

Dans $\mathbb{R}^2$:

$$Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0$$

avec $A, B, C \neq 0$.

$$A(x,y)\,\frac{\partial^2 u}{\partial x^2} + B(x,y)\,\frac{\partial^2 u}{\partial x \partial y} + C(x,y)\,\frac{\partial^2 u}{\partial y^2} + D(x,y)\,\frac{\partial u}{\partial x} + E(x,y)\,\frac{\partial u}{\partial y} + F(x,y)\,u \;=\; G(x,y).$$

	Ellipse	Parabole	Hyperbole	Coniques dégénérées
Cône ∩ plan	Page 2			
Bifocale (2 foyers)	Page 15		Page 28	
Focale (1 foyer + 1 directrice)	Page 26	Page 37	Page 36	
Etude fonctionnelle et graphique	Page 24	Page 38	Page 34	
Equation générale et réductions	Page 41			
Tangentes		Page 42		
Propriétés optique		Page 43		

	Ellipse	Parabole	Hyperbole	Coniques dégénérées
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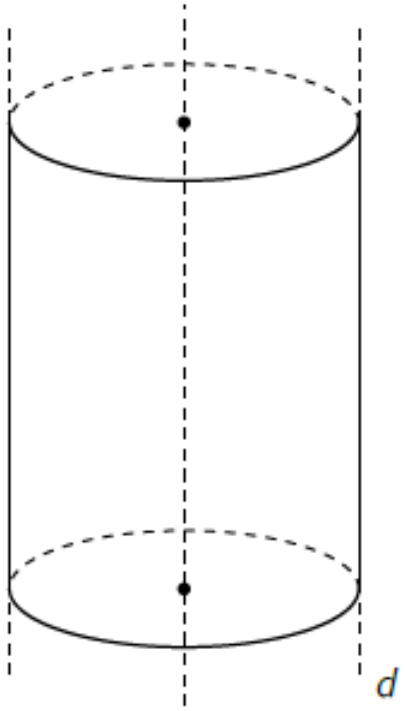
Equation générale et réductions	Page 41			
Tangentes		Page 42		
Propriétés optique		Page 43		

Ellipse

Approche bifocale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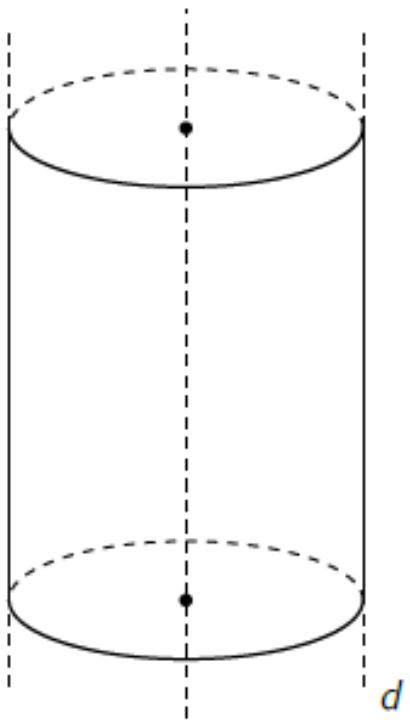
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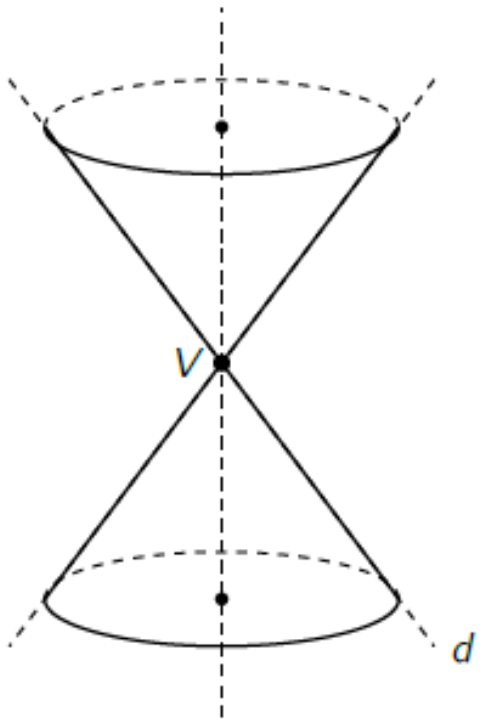
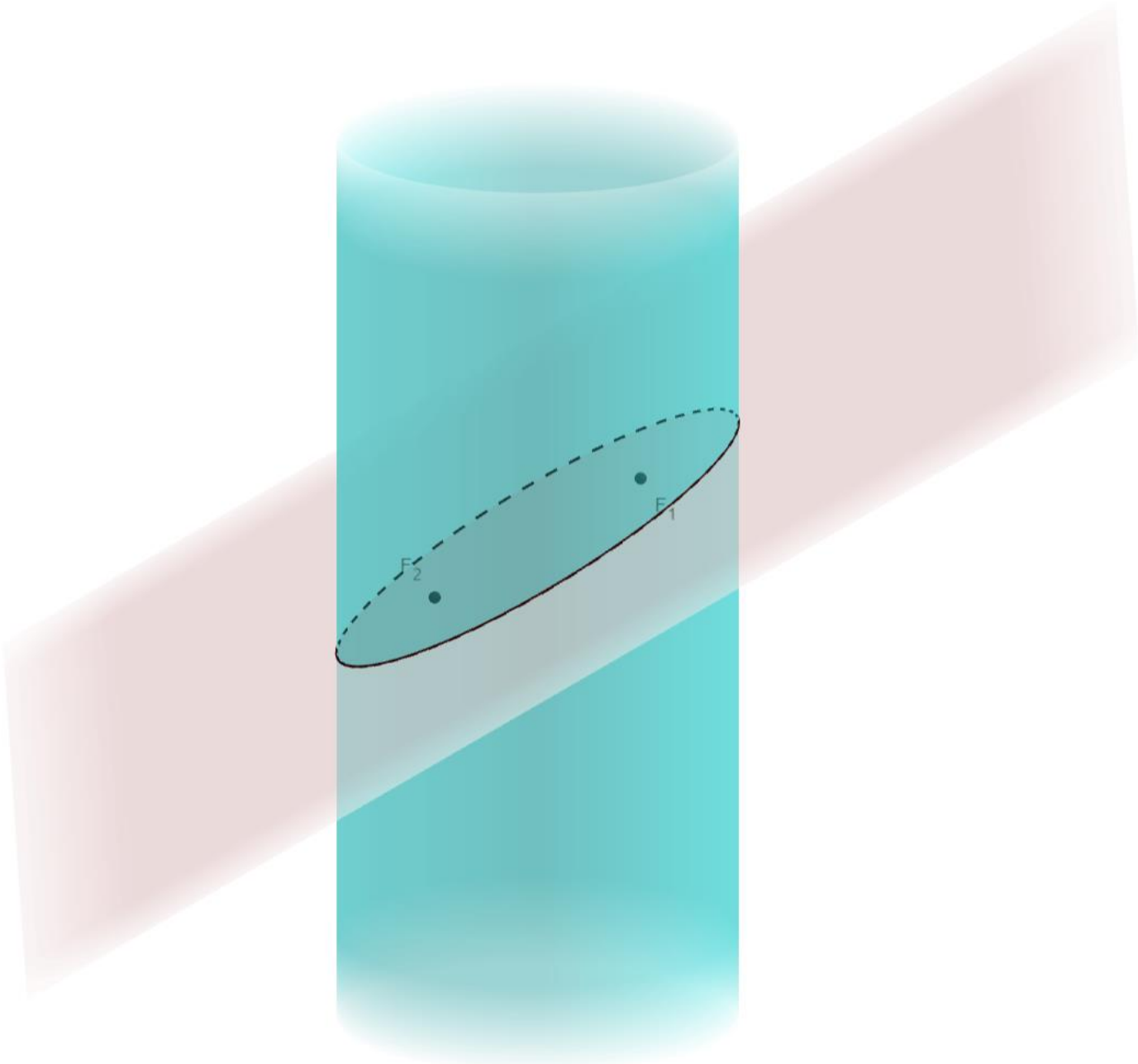
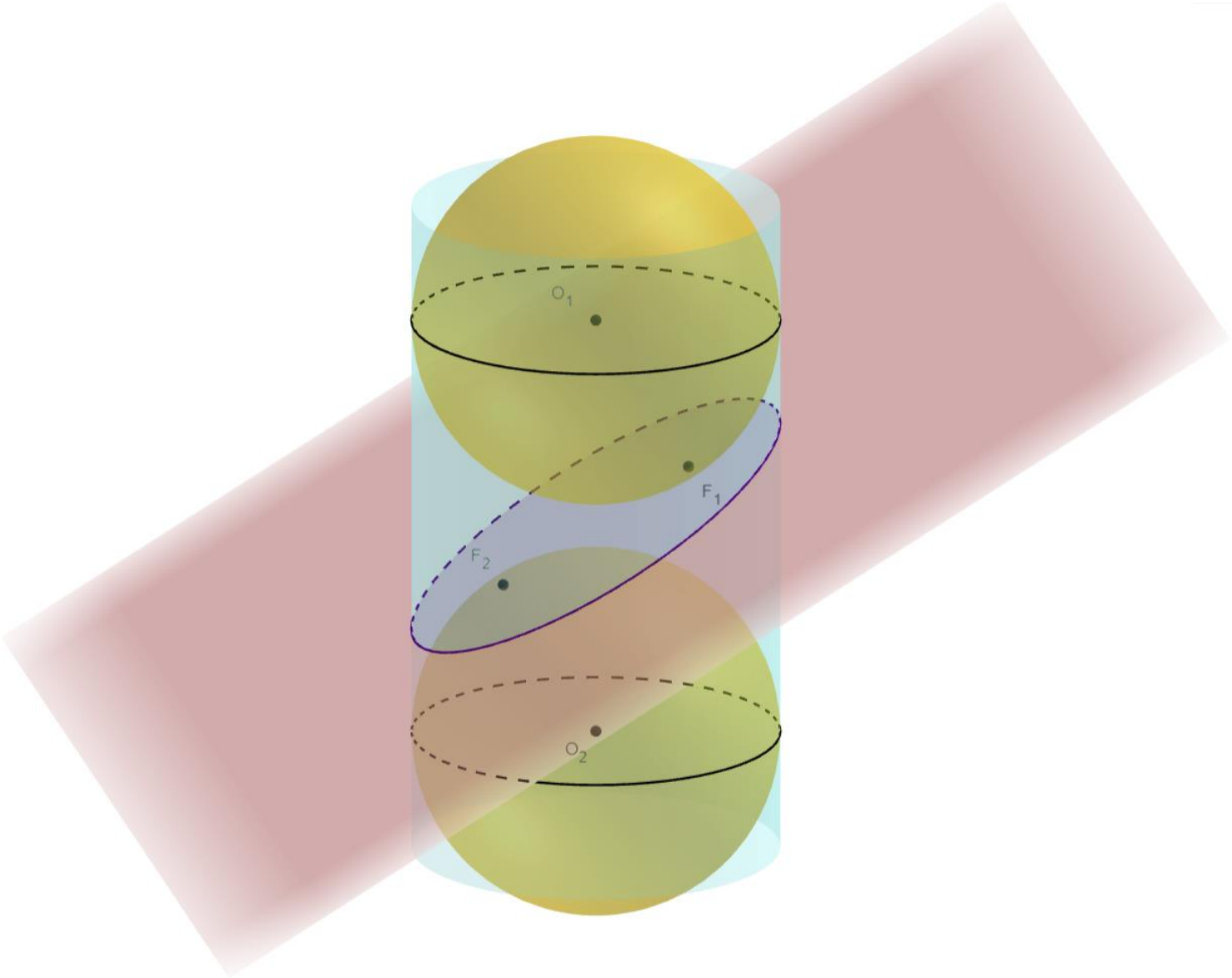
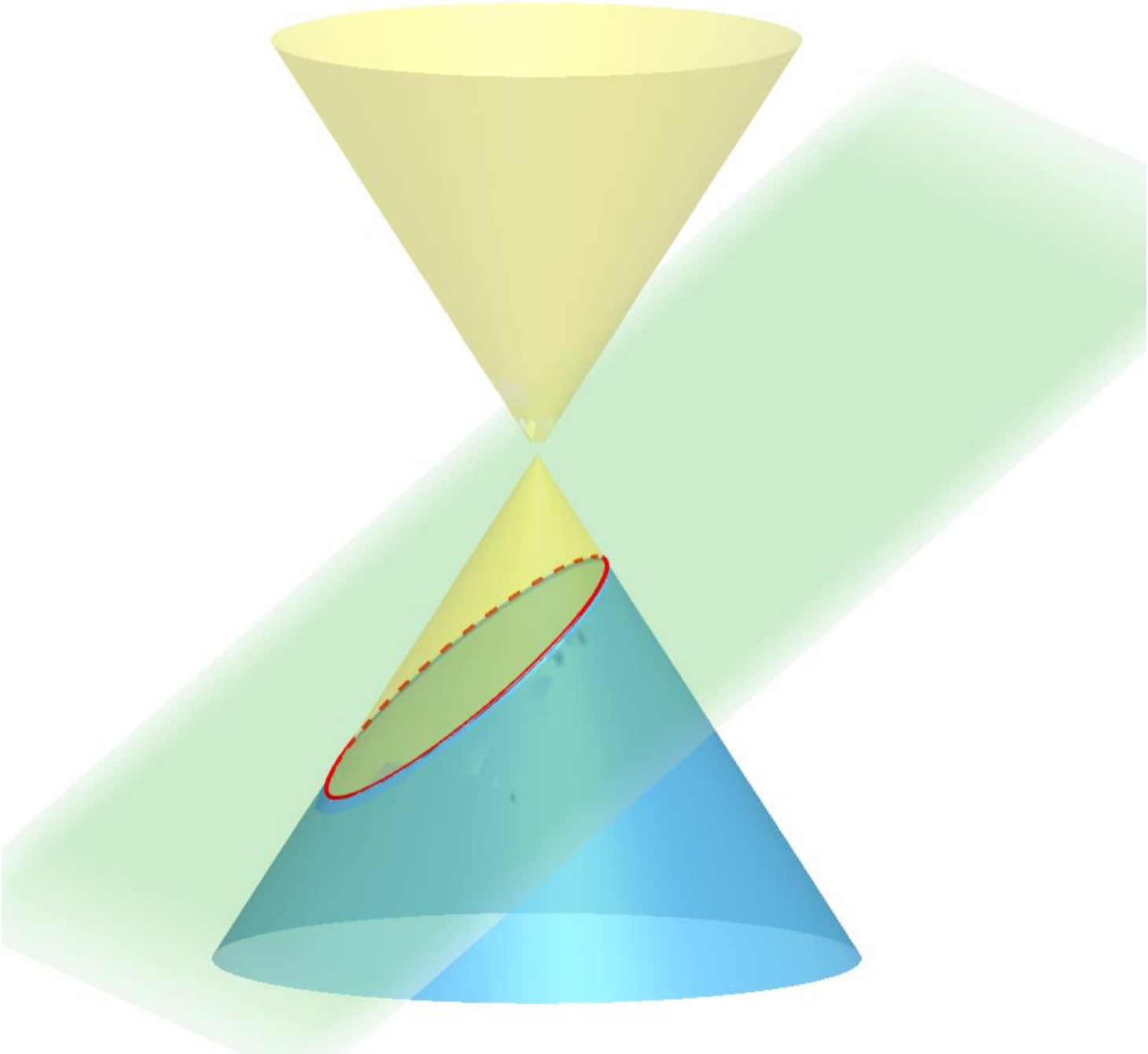
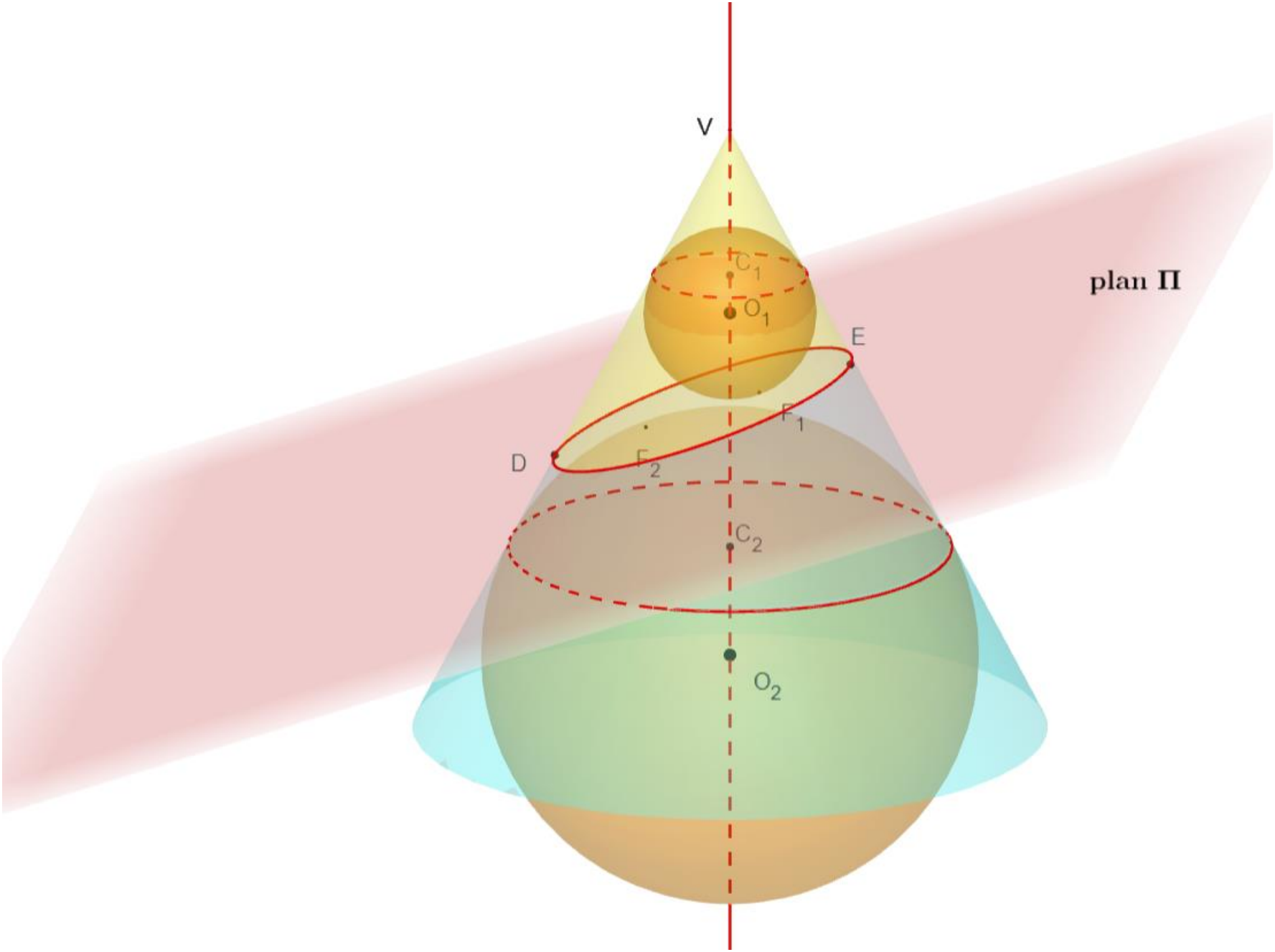

V
d


F_1
F_2



https://www.geogebra.org/classic/anmfedwx





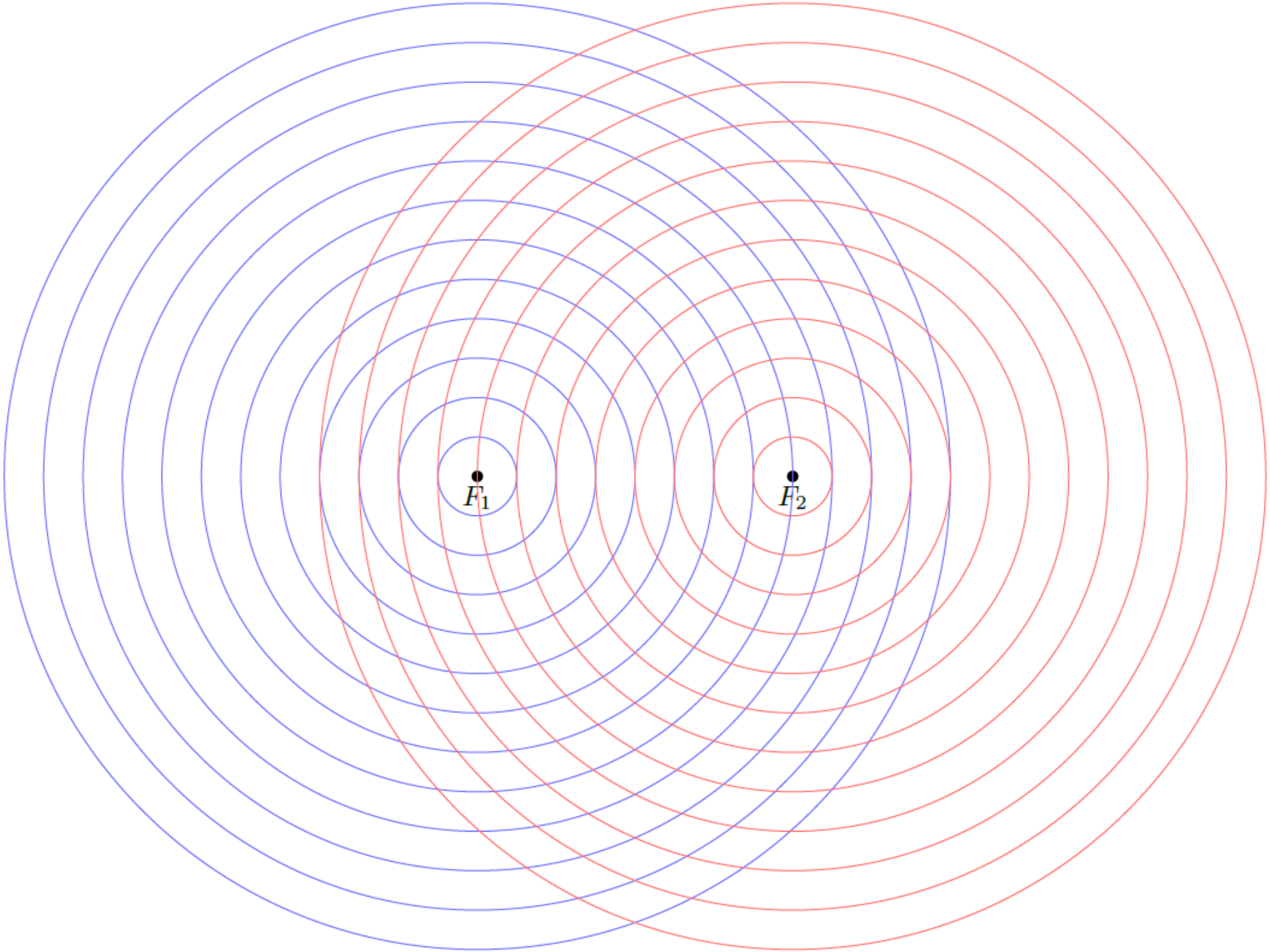
https://www.geogebra.org/classic/teehmga3

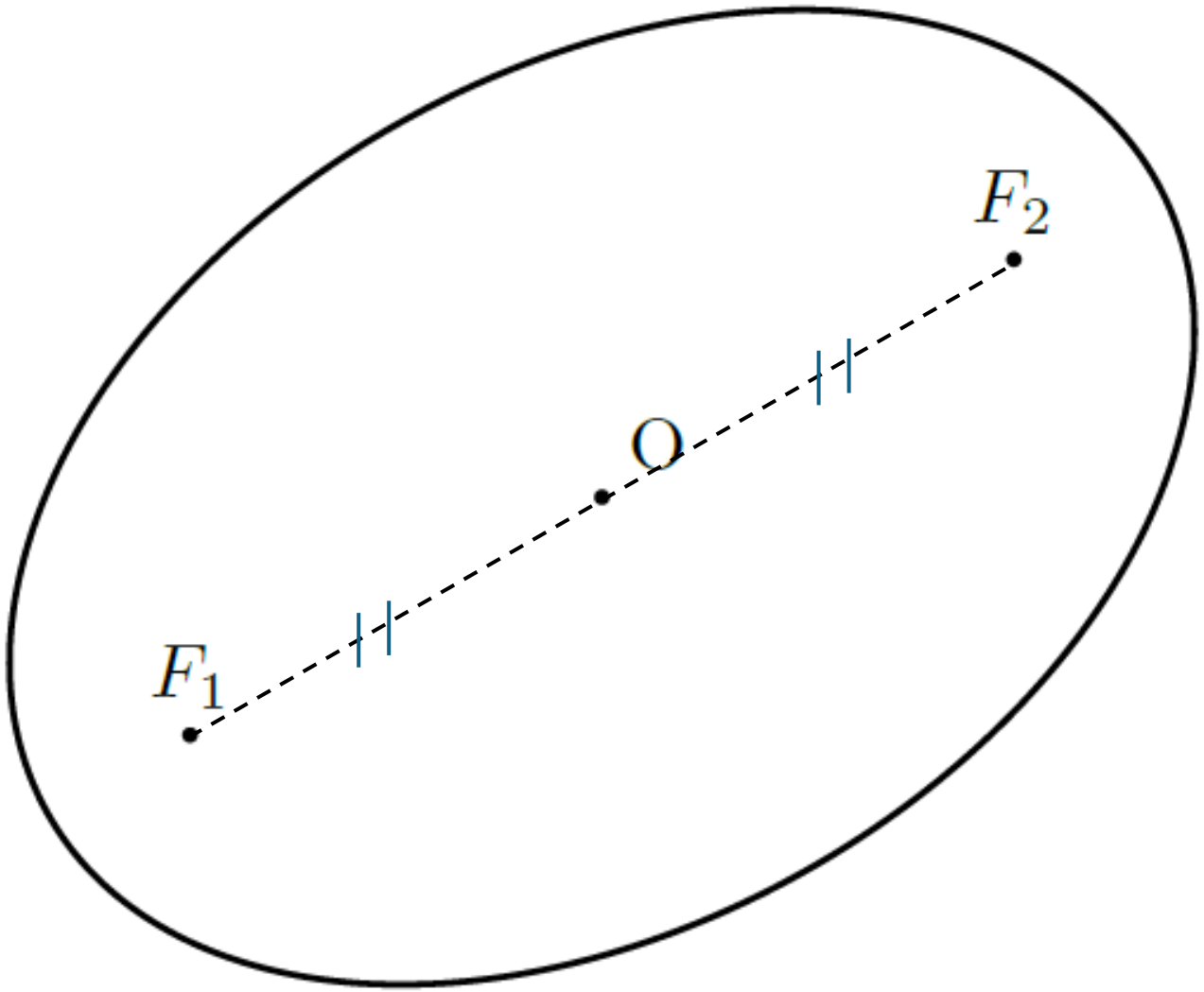
Définition bifocale de l'ellipse

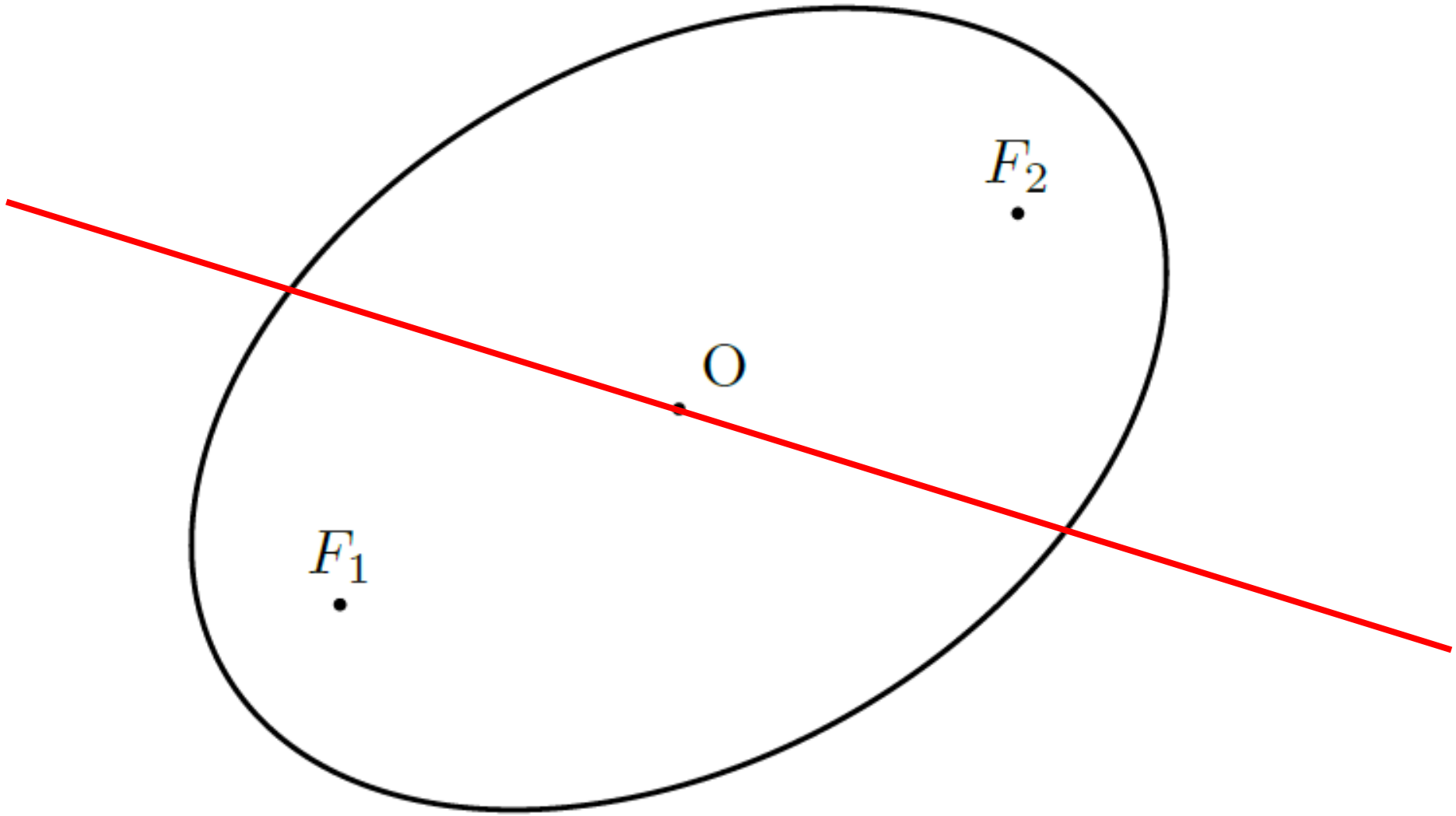
Définition 3.2. Soient F_1 et F_2 deux points distincts d'un plan Π, distants de $\overline{F_1F_2} = 2c$. Une ellipse de foyers F_1 et F_2 est le lieu des points M du plan Π tels que la somme des distances de M à F_1 et F_2 est constante, et strictement supérieure à la distance $\overline{F_1F_2} = 2c$.

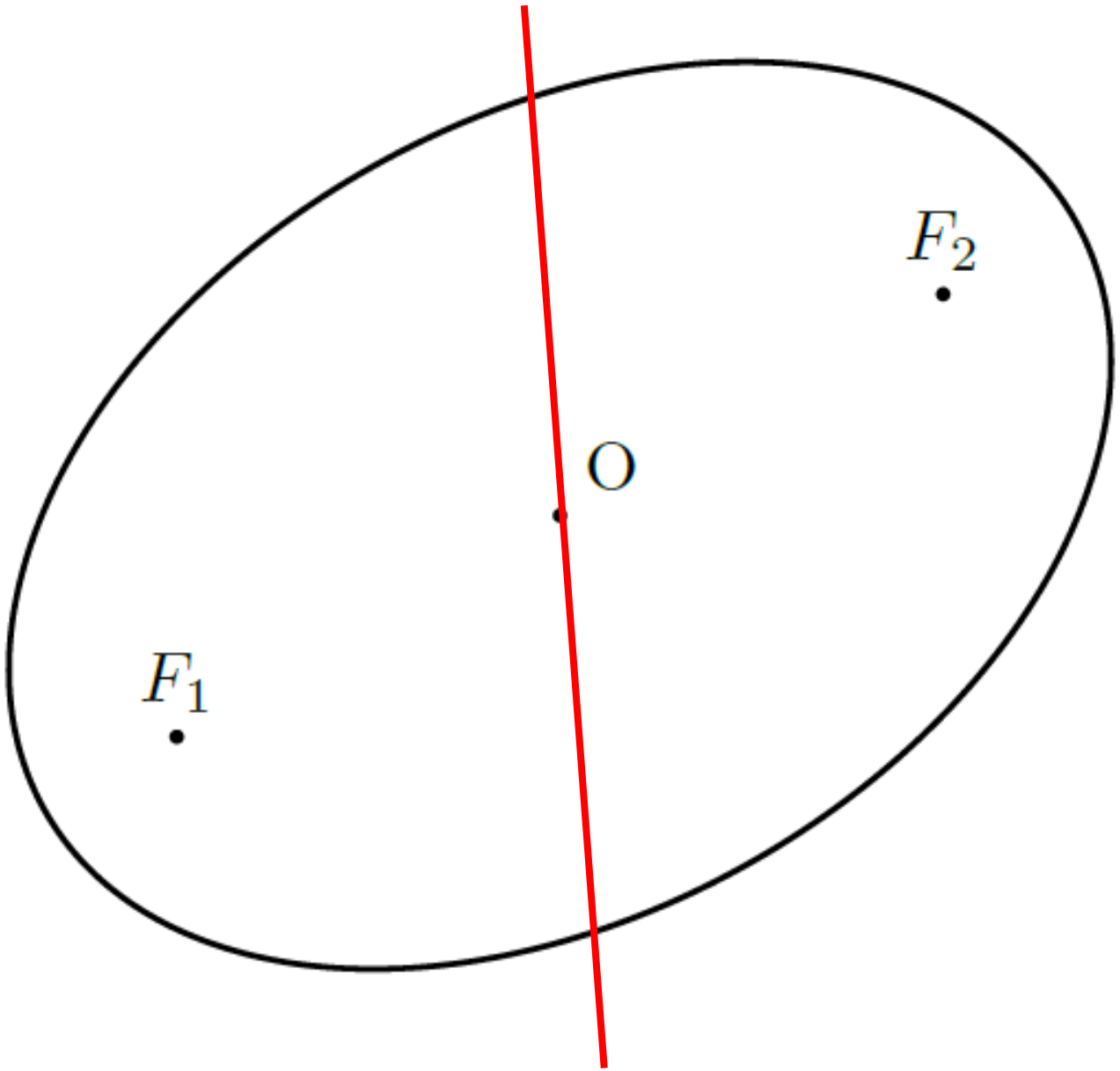
$$\mathcal{E} = \left\{ M \in \Pi \;\middle|\; \overline{MF_1} + \overline{MF_2} = 2a \right\}, \tag{3.3}$$

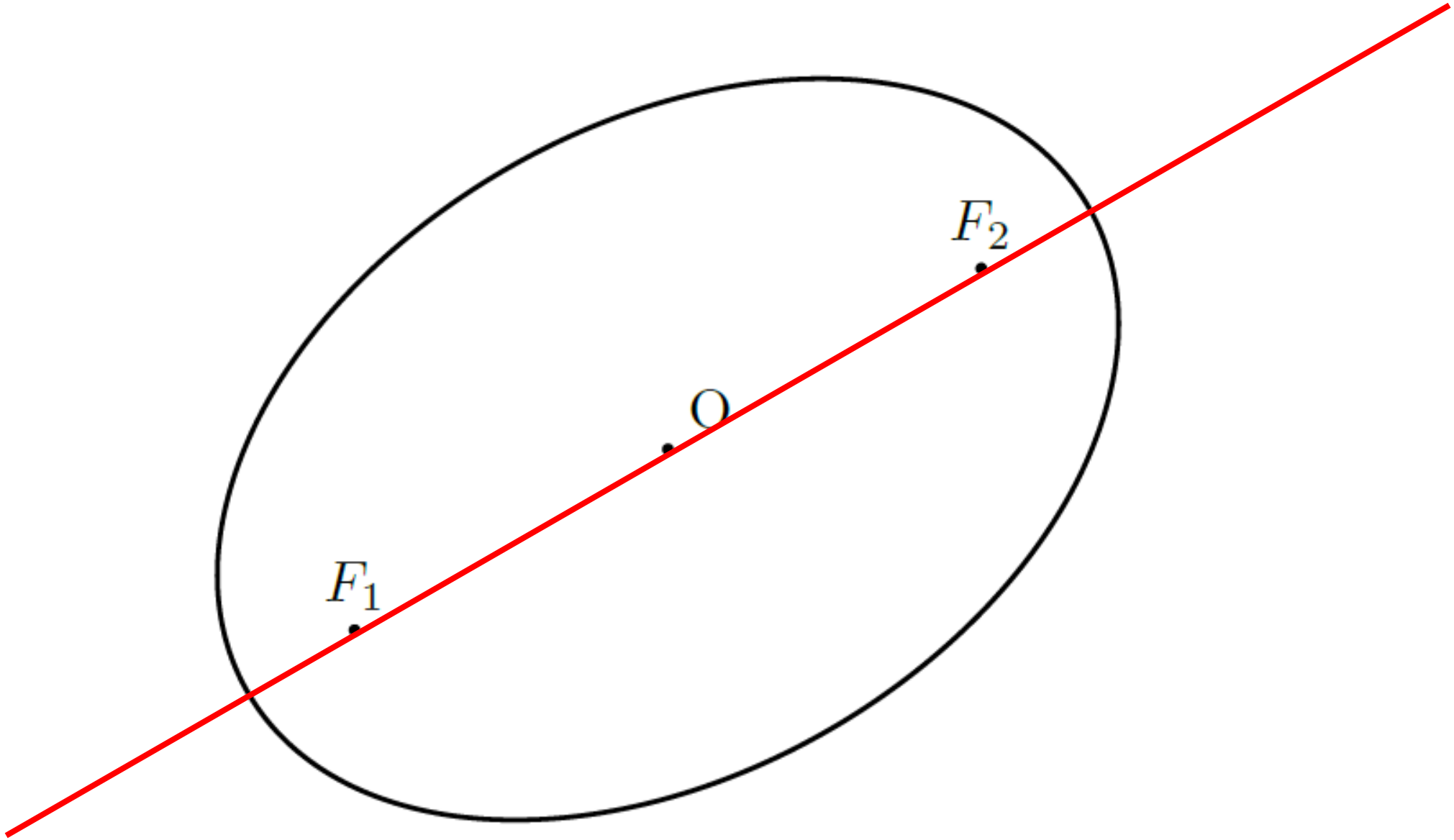
où $a > 0$ est une constante vérifiant $2a > 2c$.

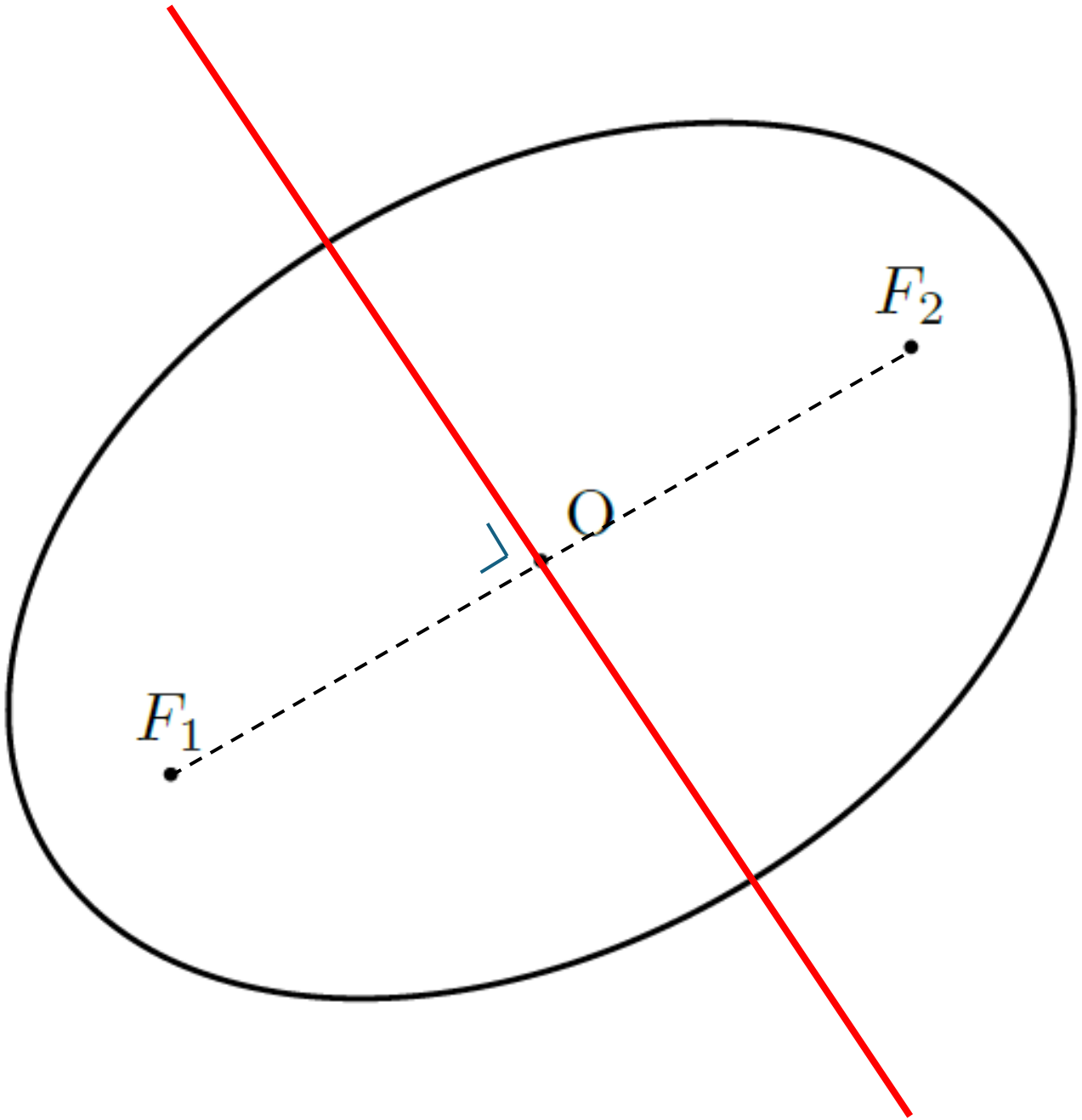

F_1
F_2

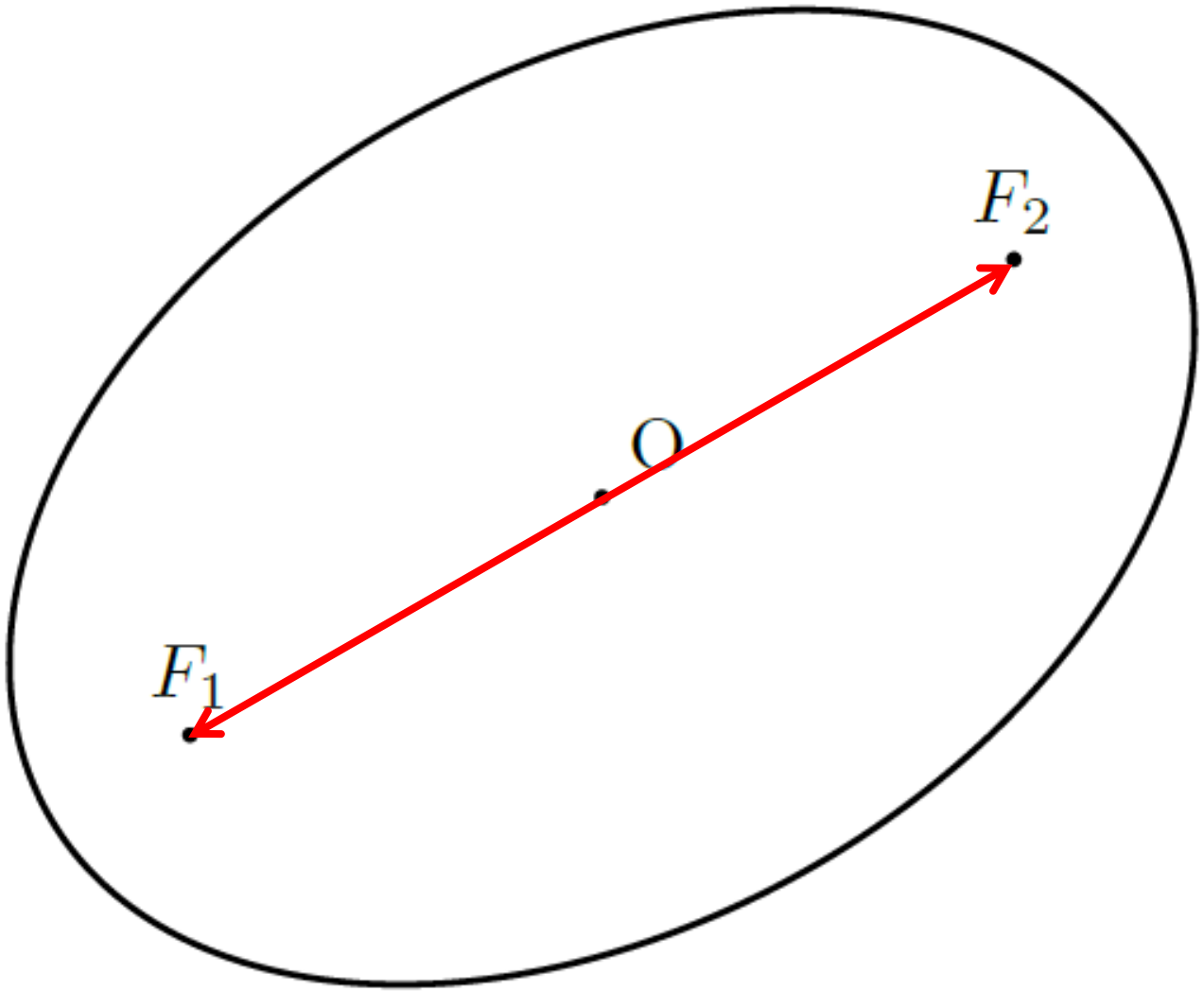

F_2
O
F_1

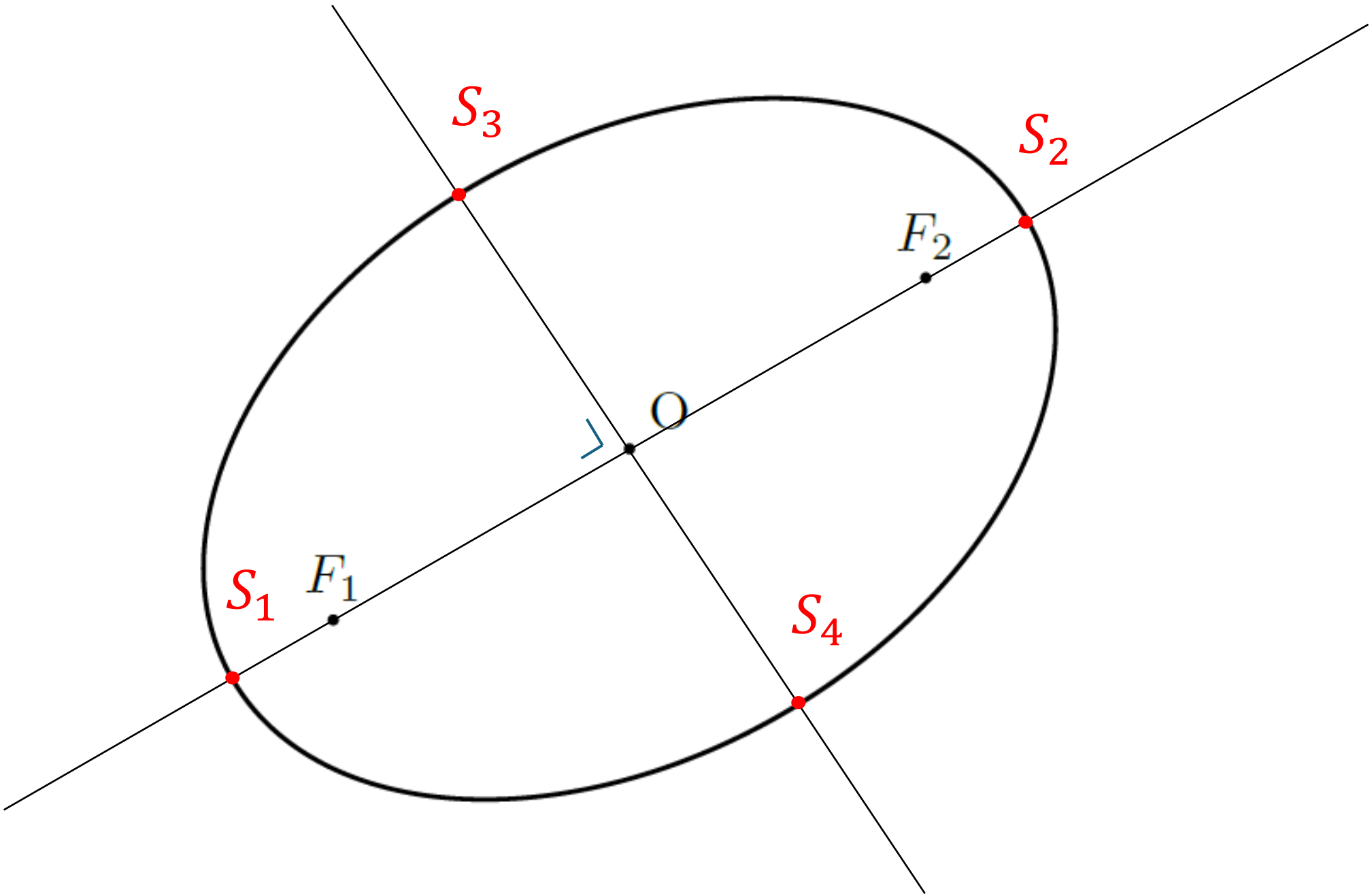

F_2
O
F_1

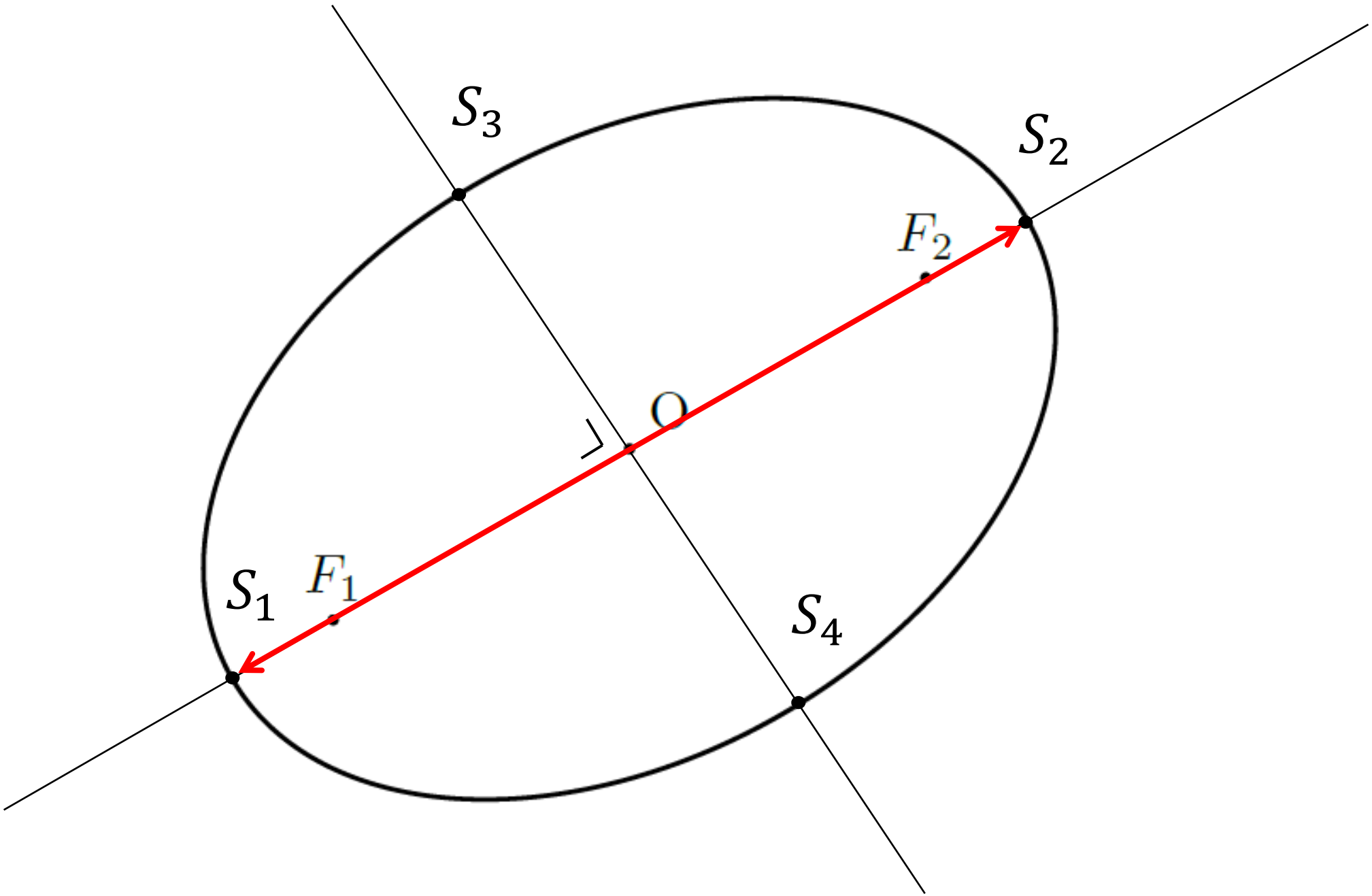

F_2
O
F_1

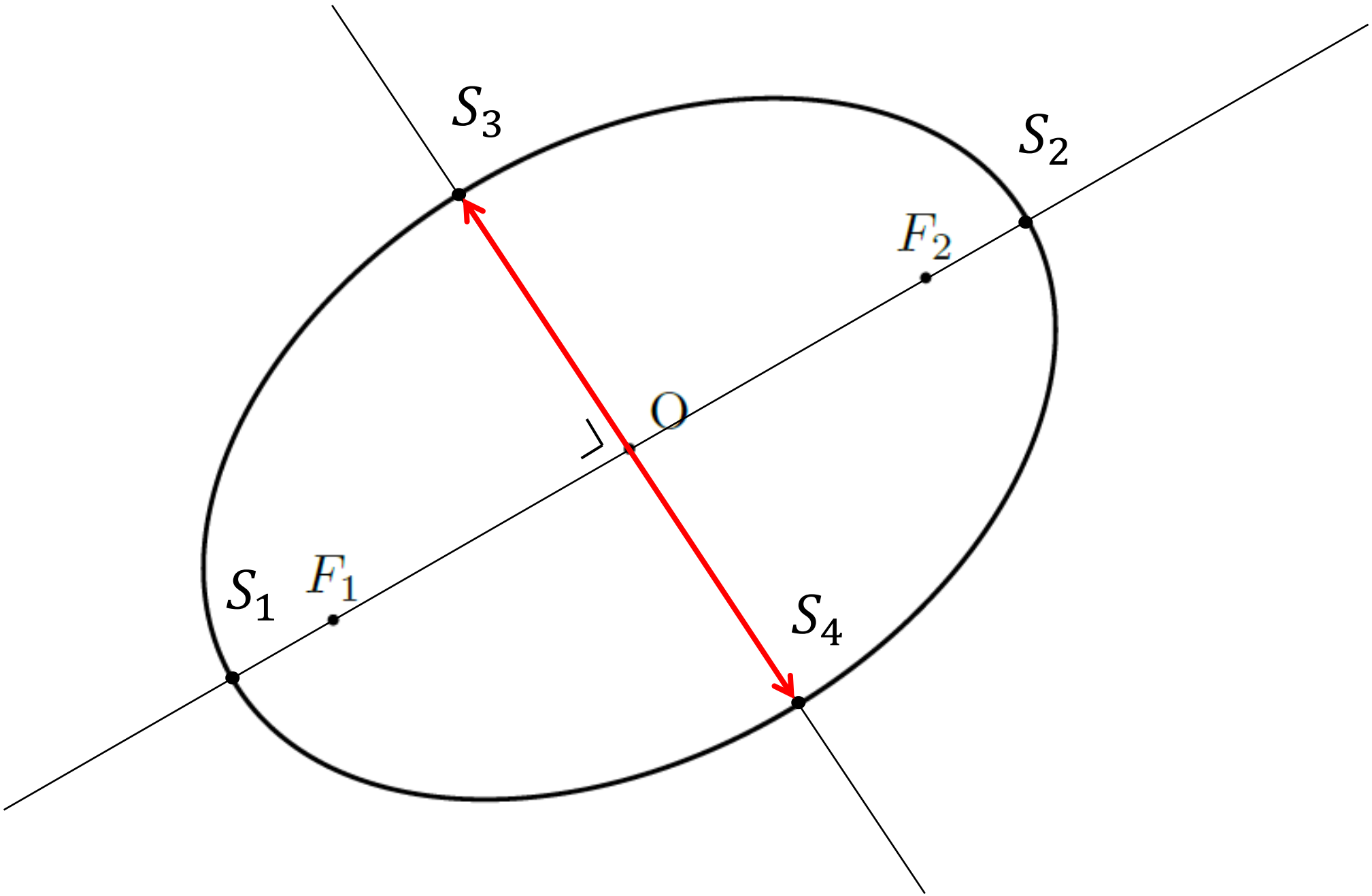

F_2
O
F_1

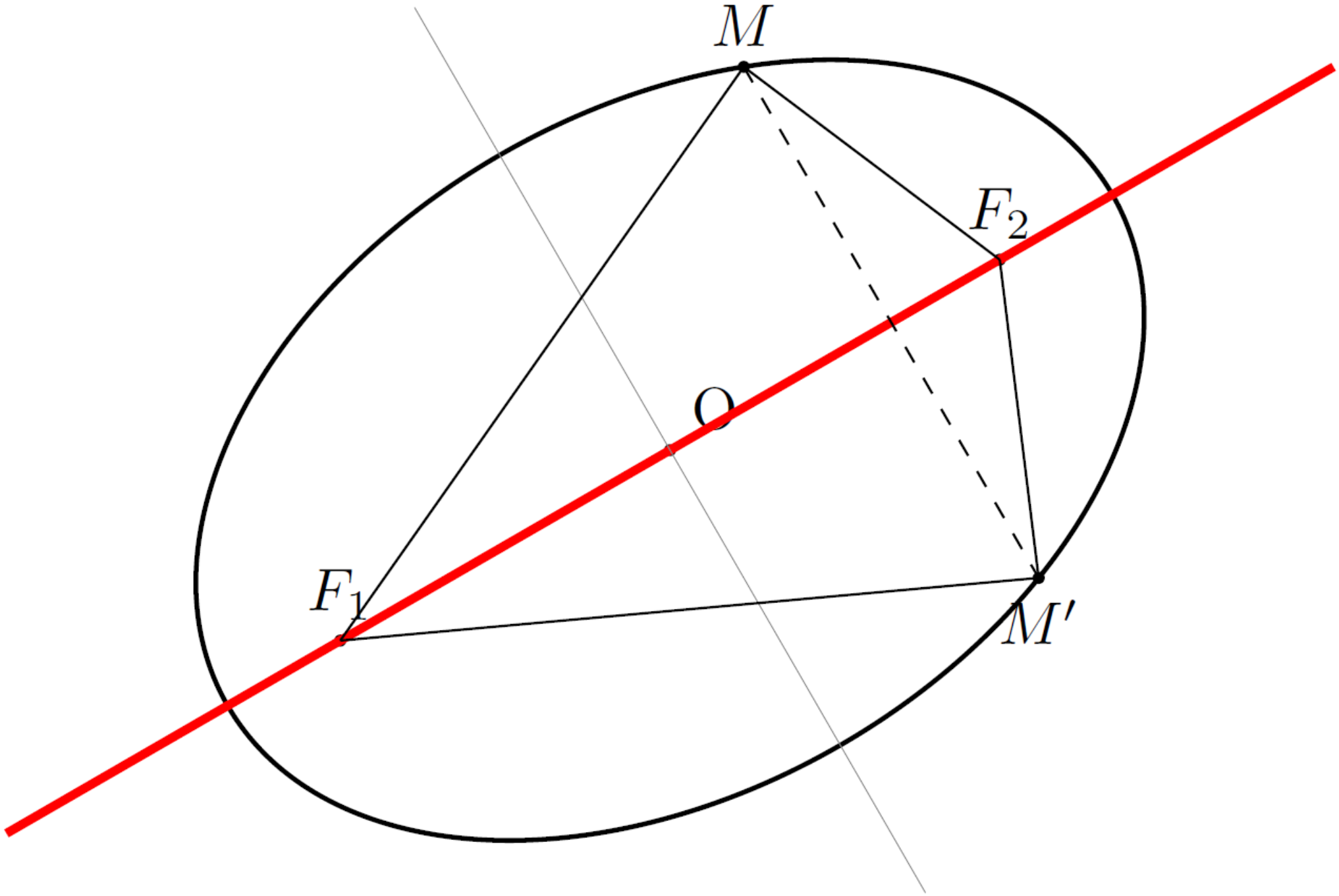

F_2
O
F_1

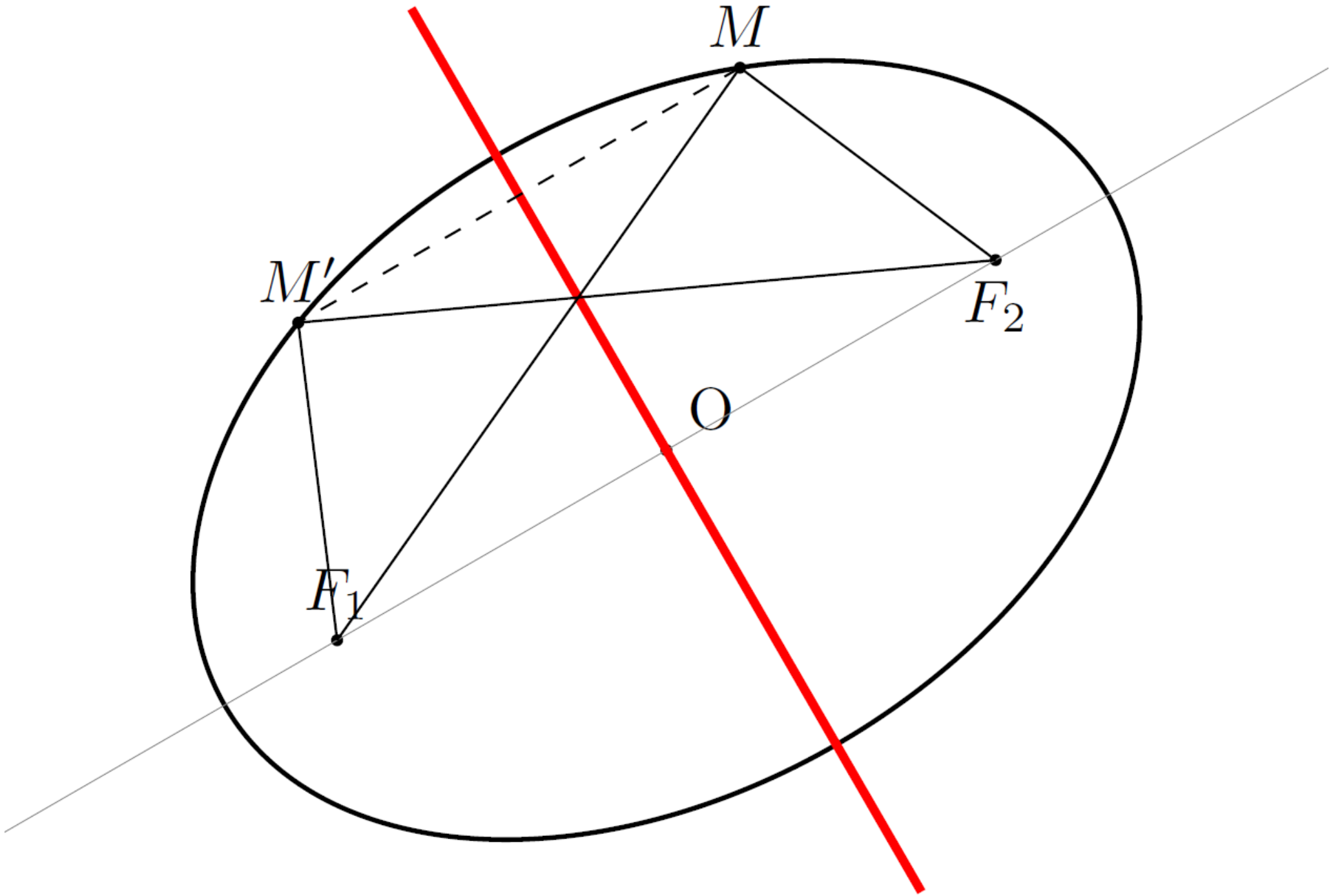

F_2
O
F_1

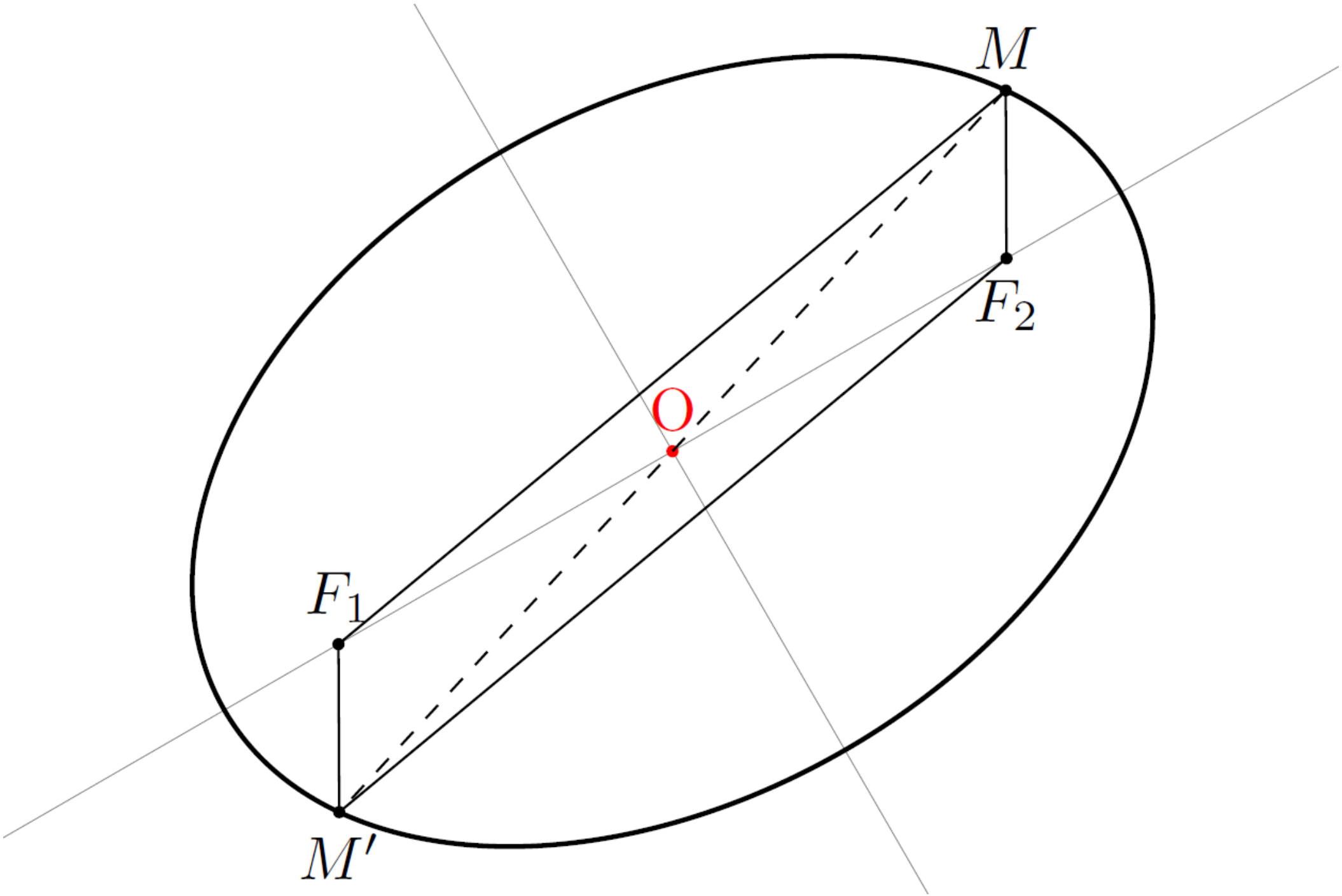

S_3
S_2
F_2
O
S_1
F_1
S_4

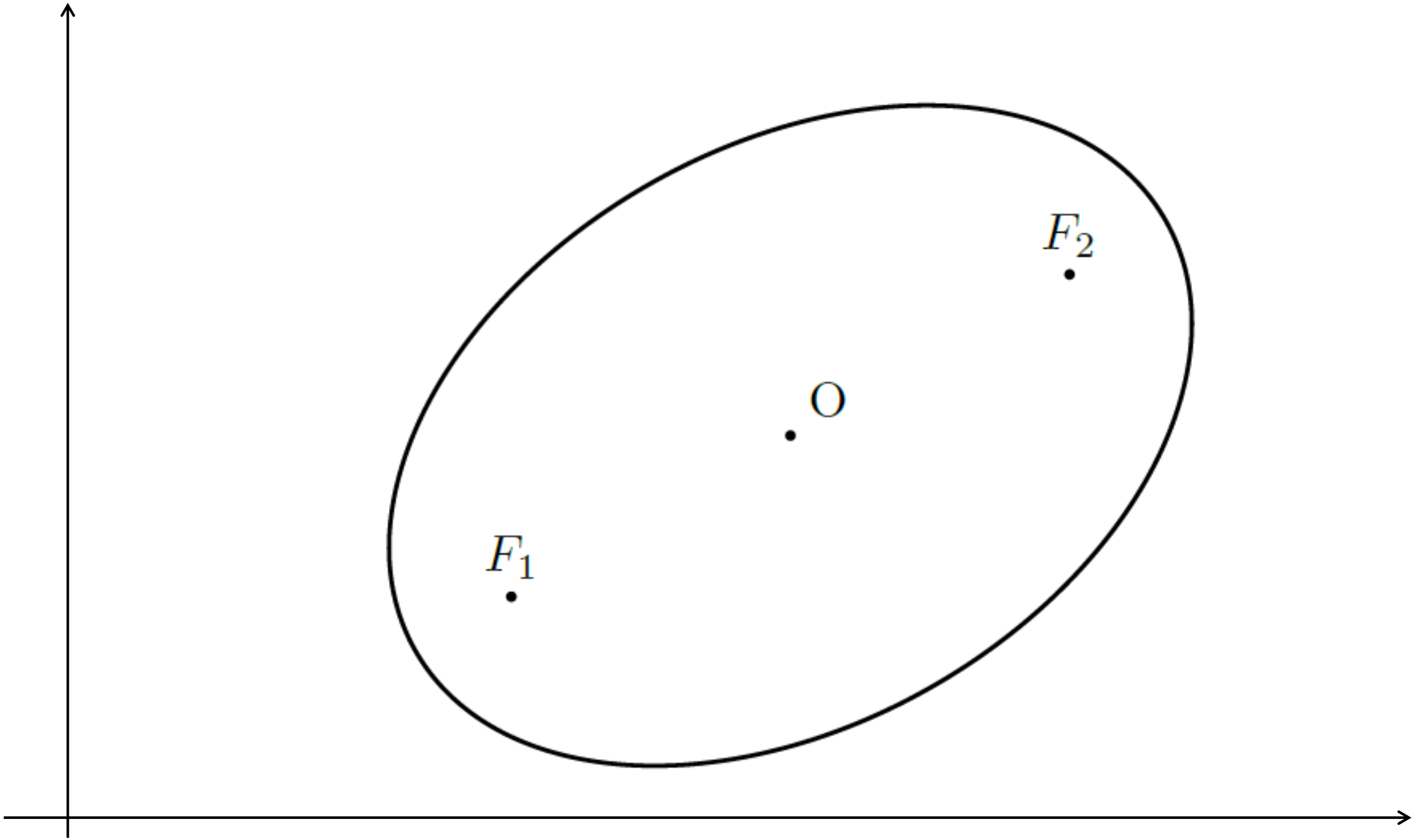

S_3
S_2
F_2
O
S_1
F_1
S_4

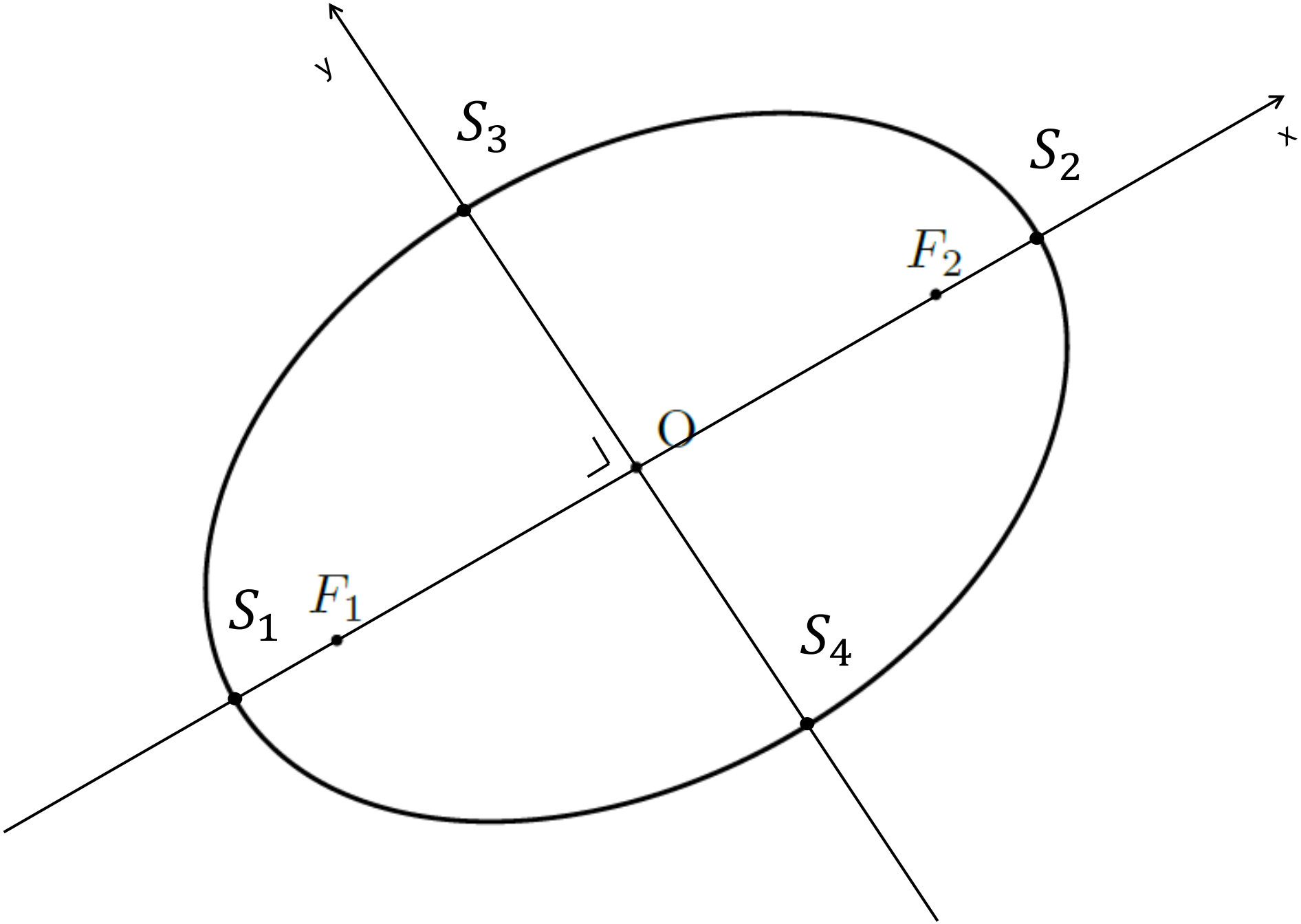

S_3
S_2
F_2
O
S_1
F_1
S_4

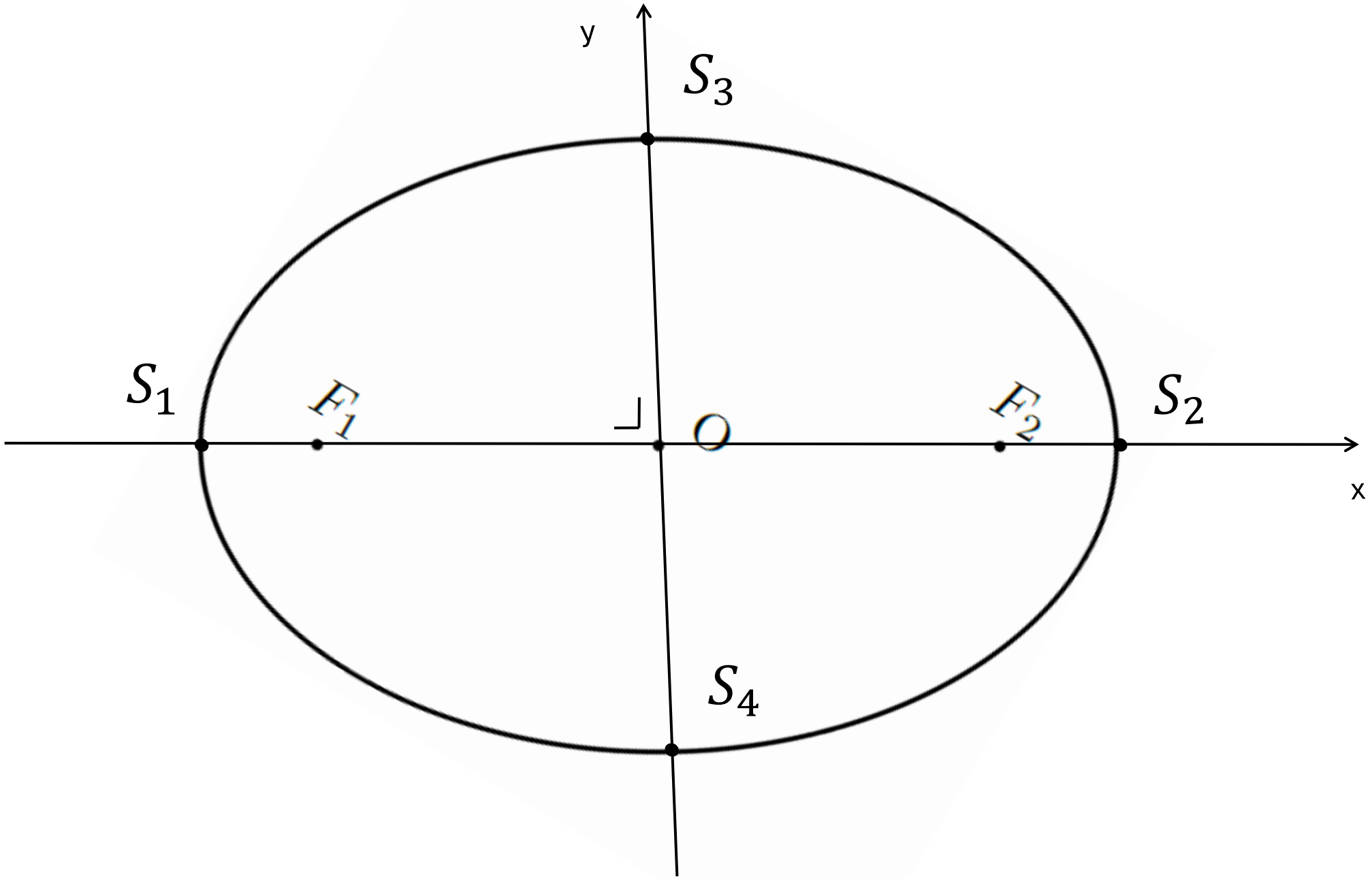

M
F_2
O
F_1
M'

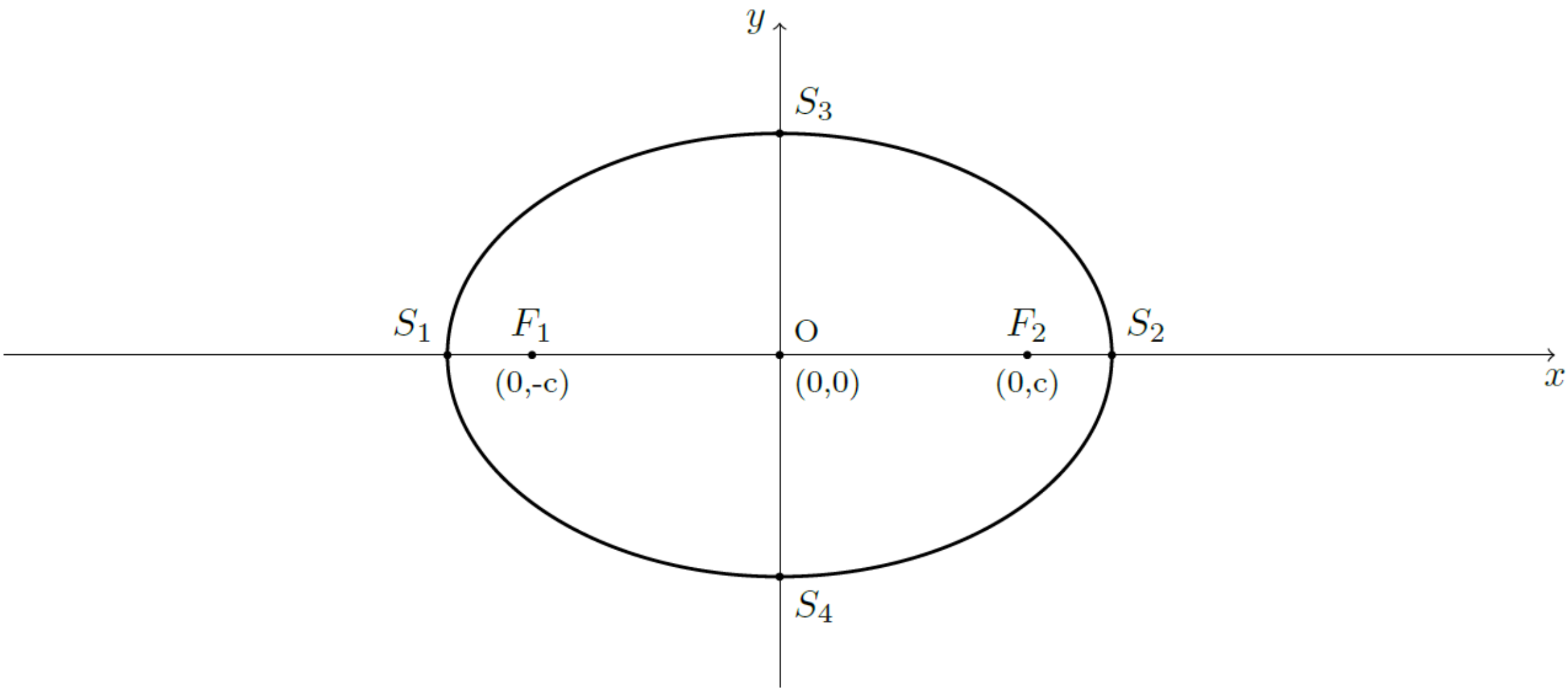

M
M'
F_2
O
F_1


M
F_2
O
F_1
M'

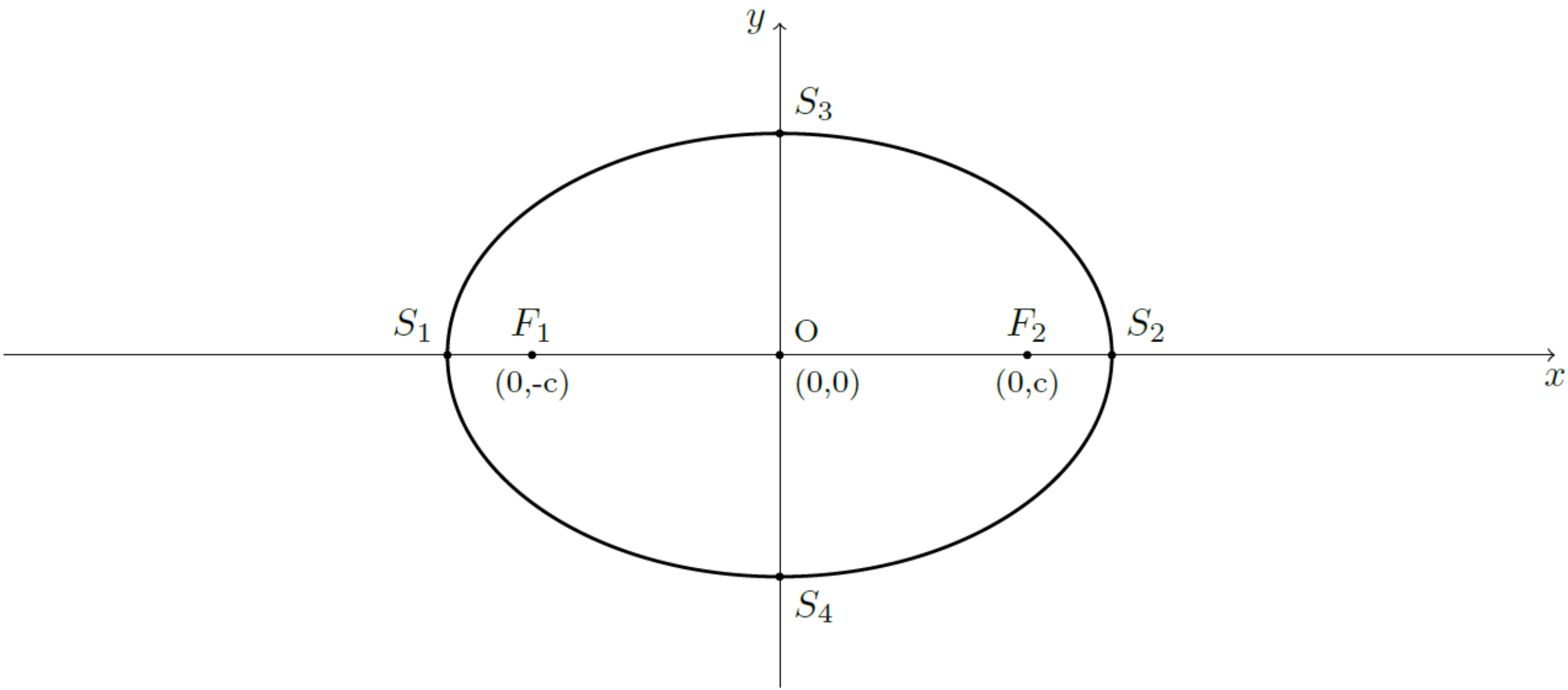

F_2
O
F_1


y
x
S_3
S_2
F_2
O
S_1
F_1
S_4

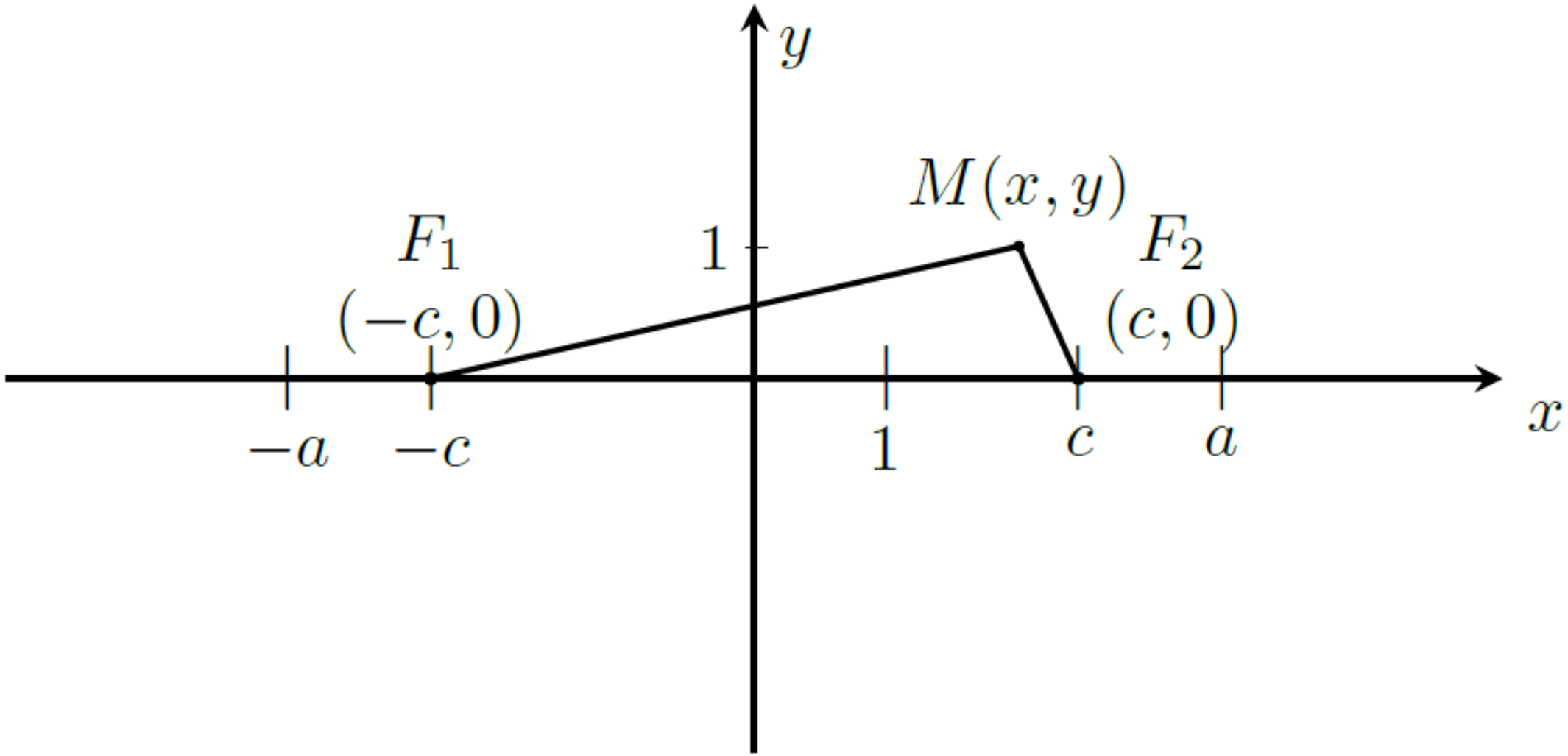

y
S_3
S_1
F_1
O
F_2
S_2
x
S_4



Exercice 3.1. Exprimez les coordonnées du sommet S_2 en fonction des longueurs connues. Déduisez ensuite les coordonnées de S_1.



Exercice 3.2. Exprimez les coordonnées du sommet S_3 en fonction des longueurs connues. Déduisez ensuite les coordonnées de S_4.



Démonstration de l'équation cartésienne de l'ellipse à partir de sa définition bifocale.

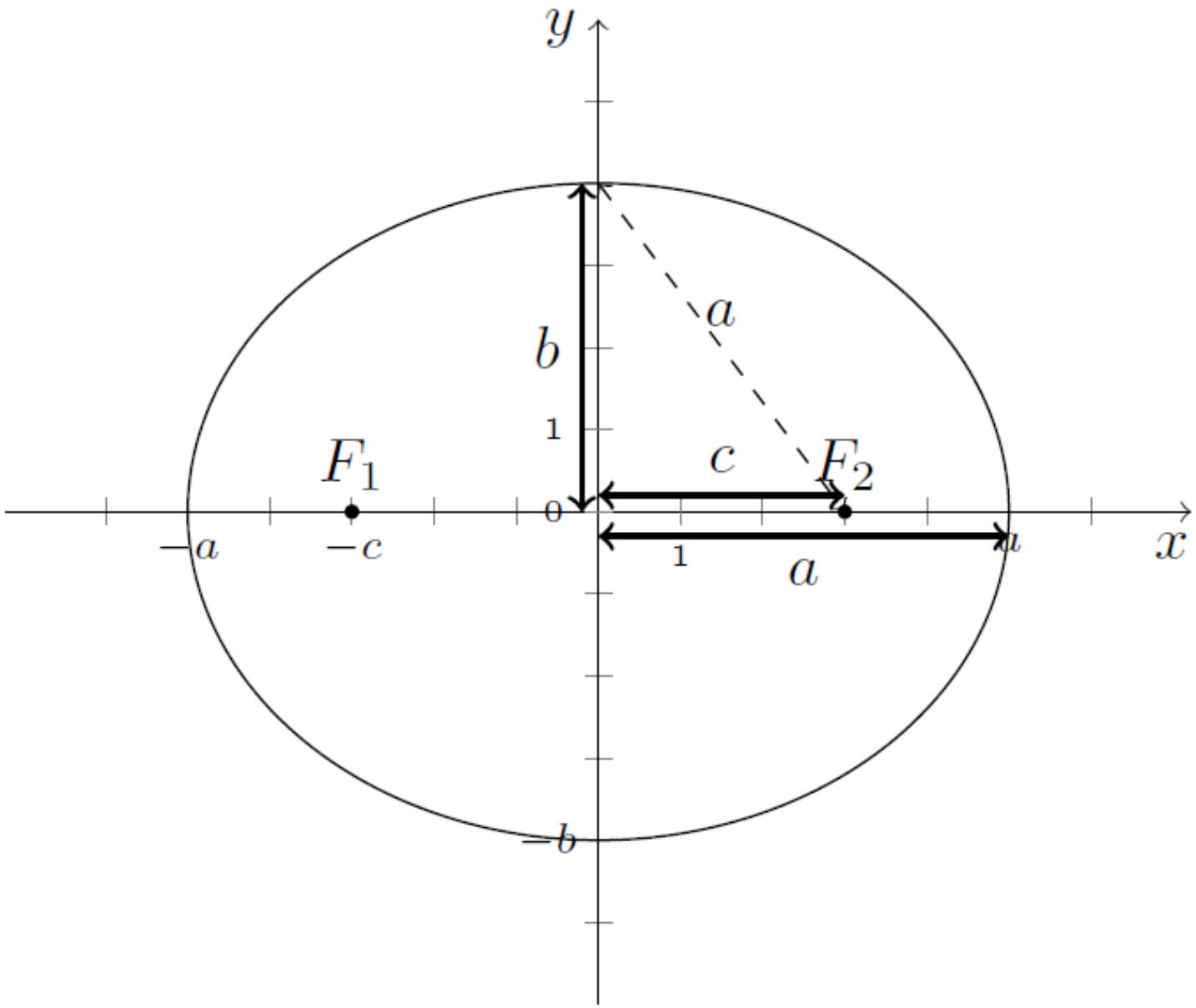
Equation cartésienne réduite de l'ellipse ou équation cartésienne rapportée à ses axes.

Soit $a, c \in \mathbb{R}_0^+$ tels que $a > c$.

Dans un repère orthonormé, considérons une ellipse dont les foyers sont $F_1(-c, 0)$ et $F_2(c, 0)$ et dont le grand axe mesure $2a$. Alors son équation cartésienne réduite s'écrit :

$$\mathcal{E} \equiv \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \tag{3.4}$$

Avec $b^2 = a^2 - c^2$


y
b
a
1
F1
c
F2
0
−a
−c
1
a
a
x
−b

Et quand ε varie ?

https://www.geogebra.org/m/aytm2czt

Et quand a et b varient ?

https://www.geogebra.org/graphing/zmu523gu